Beyond Pro- and Anti-AI: The Structure of Artificial Intelligence Attitudes in Mass Publics*

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Abstract

What is the structure of mass attitudes toward artificial intelligence (AI): which attitudes are distinct, how do they connect once the rest of the belief system is held constant, and which groups hold which combinations? To answer these questions, we conducted exploratory factor analysis on a nationally representative sample of US adults, analyzing the covariance of over fifty items across three AI applications: algorithmic decision-making and content moderation, facial recognition, and driverless vehicles. Three findings characterize the structure. First, ten factors separate by the social context of use rather than by technology (facial recognition alone splits into four factors), with self-reported AI awareness emerging as a distinct knowledge dimension. Second, the partial-correlation network among the factors has clear structure, but within this system no factor functions as a general-attitude hub. Third, the same social cleavages move different dimensions in opposing directions: older Americans are less optimistic about algorithmic and driverless-vehicle applications yet more accepting of facial recognition, and partisanship structures content-moderation attitudes (adjusted $R^2 = .33$) while leaving public-surveillance acceptance nearly unstructured ($R^2 = .04$), so a single general scale would average away the cleavages that matter most. Industry and policymakers need to recognize this architecture to develop technologies and regulations that are effective and democratically accountable, in line with public sentiment toward AI.

Keywords: artificial intelligence, public opinion, political attitudes, belief systems, exploratory factor analysis, psychometric network, survey methodology

1. Introduction

Artificial intelligence (AI) applications have rapidly proliferated in today's economy and social life. Use cases and domains where a given family of AI technologies is deployed vary widely, including entertainment, education, healthcare, transportation, the military and law enforcement (Anantrasirichai and Bull, 2022; Baidoo-Anu and Ansah, 2023; Secinaro et al., 2021; Berk, 2021; Abduljabbar et al., 2019; Szabadföldi, 2021). In turn, people's experiences and perceptions influence the degree of popular support and adoption of these technologies (Isaksen et al., 2025; Dorotic et al., 2024). Thus, as elected representatives and multilateral institutions observe this cycle, public views on the uses of AI technologies in society can affect policy attention and regulatory efforts (Schiff, 2024; European Commission, 2021).

As a result, measuring and studying public opinion on AI has become an emerging interest among scholars at the intersection of society, politics, and technology (Araujo et al., 2020;

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Brauner et al., 2023). While studying citizens' opinions and perceptions of technology has a long tradition (Slovic, 1987; Davis, 1989), AI raises specific measurement questions because of the technology's general nature (Schepman and Rodway, 2020; Grassini, 2023). If the internet and social media sparked debates about sharing and disseminating information (boyd and Ellison, 2007), and nuclear energy conversations centered on its economic benefits as well as environmental and health risks (van der Pligt et al., 1986), we think AI has a much broader and more diffuse scope of influence. AI applications and technology vary greatly, from autonomous physically embodied machines to recommendation algorithms and classification systems that use image recognition (Hadlington et al., 2023; Araujo et al., 2020; Brauner et al., 2023). Overall, a single AI technology can reach multiple application domains (Bommasani et al., 2021), since it can often be used in medical, educational, law enforcement, and entertainment settings, among others.

We distinguish between two distinct questions relevant to research on AI attitudes. The first is whether citizens hold a broadly favorable or unfavorable orientation toward AI as a general object (Schepman and Rodway, 2020; Stein et al., 2024). The second is how citizens' evaluations of specific AI applications are organized: which evaluations go together, and which separate by application and context of use. Both questions matter, but this paper addresses the latter.

A substantial validation literature has produced general-attitude instruments, including the General Attitudes towards Artificial Intelligence Scale (GAAIS) (Schepman and Rodway, 2020), ATTARI-12 (Stein et al., 2024), the ATAI scale (Sindermann et al., 2021), and AIAS-4 (Grassini, 2023). These scales measure the extent to which people have a generally positive or negative orientation toward AI as a general object. GAAIS, for example, asks respondents how much they agree with statements such as "Artificial Intelligence is exciting" and "I think Artificial Intelligence is dangerous" (Schepman and Rodway, 2020, Table 3). In contrast, we examine how citizens' opinions are organized when they evaluate concrete AI applications in specific social contexts. Our focus is on describing the structure of attitudes about specific AI applications, and does not include a multi-item general-orientation scale. Nevertheless, this analysis is important because if attitudes are organized by domain and context, a given sociodemographic or political predictor may relate to different applications with varying strength and sometimes opposite sign. If so, a single index built from attitudes across all applications would hide those differences.

The question of how citizens organize attitudes about political objects has a long tradition in the study of mass belief systems. Converse (1964) defined a belief system as a configuration of ideas and attitudes bound together by constraint, and found that the issue positions of typical citizens were only weakly correlated. Recent work models such systems directly as networks, with attitudes as nodes and their interrelationships as edges (Brandt et al., 2019; Brandt and Sleegers, 2021). Unlike spatial models of ideology, which reduce attitudes to a few latent dimensions, this approach makes few assumptions about how attitudes ought to relate to one another (Fishman and Davis, 2022), and it treats constraint as an empirical property of the network rather than a definitional one (Brandt and Sleegers, 2021). Survey research on AI governance has found that Americans distinguish sharply among the actors who develop and manage AI, placing far more confidence in university researchers and the military than in Facebook (Zhang and Dafoe, 2020). Whether evaluations of the applications themselves cohere into a single disposition or divide by context of use is what a factor structure and its network can show.

In the present study, we explore the structure of AI attitudes. We ask the following questions: Are the attitudes based on a *single* underlying dimension (e.g., a simple positive versus negative attitudinal space), or are they *domain*-specific (e.g., personal assistants, autonomous vehicles,

face recognition, and so on)? What is the role of the social and political *context* surrounding those domains, and could AI attitudes, even within a single domain, differ depending on the specific application of the technology? Third, how do the distinct attitudes *connect*: once the rest of the attitude system is held constant, is there a general core through which the dimensions relate? Fourth, do social and political *cleavages* align with a single pro-anti axis, or do they cut across the attitude dimensions? Our substantive goal is to identify such structure based on empirical evidence and suggest a theoretical foundation for the study of AI attitudes. Our second goal is methodological: we draw implications from this structure for how surveys measure AI attitudes.

Defining "artificial intelligence" is a challenge that evolves with the technology's advancement and adoption (Monett and Lewis, 2018; Wang, 2019); for our purposes, AI involves the design and application of machines that learn and apply techniques to achieve goals in dynamic environments (Rajaraman, 2014; Auernhammer, 2020). AI could be compared, simplistically, to a tool like a hammer for a carpenter, but such an analogy does not account for the societal and political ramifications that distinguish one application of the technology from another.

Among US adults surveyed in late 2021, the 52 items we analyze resolve into ten correlated factors that separate by the social context in which AI is used. Facial recognition alone splits into four factors (acceptance of police use in public spaces, acceptance of commercial uses, judgments of how well it performs for police, and anticipated surveillance harms), content moderation separates concern from efficacy, and self-reported awareness forms a distinct knowledge dimension. The factors are correlated, but none serves as a central hub linking the rest, and any general factor above them is weak. The same social groups also take opposing positions across applications. Americans aged 65 and older, for example, are less optimistic than those aged 18–29 about practical uses of AI but more accepting of commercial facial recognition. Combining all 52 items into a single index hides much of this variation: the age gap shrinks, and the ideological and Black–White differences found on specific factors disappear.

Whether AI should be governed by universal rules or application by application depends on whether public opinion is organized by application. Public attitudes toward AI plausibly draw on many considerations, from perceptions of bias and discrimination to concerns about privacy and autonomy to trust in the institutions that develop and oversee the technology. Which of these considerations cohere into distinct attitude dimensions, and for which applications, is part of what a structural analysis can show empirically.

2. Study Design and Analysis

We use data from Pew Research Center's wave 99 of the American Trends Panel, a survey of 10,260 U.S. adults conducted between November 1st and 7th, 2021, to identify a structure of attitudes toward artificial intelligence. The survey contains questions about general attitudes toward artificial intelligence, devoid of a particular domain, and about three applications: (a) facial recognition technology and surveillance, (b) algorithms that social media companies use to detect misinformation, a module that ends by asking whether such programs should make final decisions about mortgages, medical treatment, job applicants, and parole, and (c) driverless passenger vehicles. A subset of the respondents (N = 5,153) answered questions about artificial intelligence, while the other half were asked about other technologies such as gene editing and exoskeletons.

The survey was fielded a year before ChatGPT's public release. These data therefore record how the public organized its evaluations of AI applications before generative AI became a shared public referent for "AI", and they are a pre-generative-AI baseline. They are not a picture of current opinion, and the Pew battery does not cover every relevant domain.

We employ exploratory factor analysis (EFA) to explore a potential structure of beliefs about artificial intelligence using the set of survey questions about AI from the American Trends Panel. EFA is a flexible method that enables researchers to identify latent factors explaining the patterns of correlations within a set of observed variables. By grouping related variables, EFA simplifies data and reveals underlying dimensions (Costello and Osborne, 2005). For instance, in a study examining attitudes toward technology, EFA can show whether questions about technology usage relate to specific factors, such as ease of use or perceived usefulness. Attitudes toward AI, irrespective of domain, may be explained by a single positive-negative dimension. Conversely, respondents' beliefs about AI might not converge on a universal set of factors but rather be influenced by specific domains (e.g., personal assistants, military technologies, healthcare) or the political context, such as the significance of government regulation. The structure identified by EFA is specific to the dataset used to estimate it. Examining whether the identified structure applies to another sample requires confirmatory factor analysis (CFA) on the new dataset (Brown, 2015).

For the exploratory factor analysis (EFA), we used a set of 52 survey questions (see Table B.2 in the Appendix). The analytical process categorizes these questions based on the correlation patterns among the responses. For instance, if participant responses to all items concerning AI in facial recognition correlate highly, these items would form a "facial recognition" factor, potentially along a positive-negative dimension. On the other hand, if the data reveal distinct domains within the public's perception of facial recognition technology, a multi-factor model would capture the observed variance better than a single factor model. Because the survey items are ordinal, the factor analysis is estimated from the matrix of pairwise polychoric correlations rather than Pearson correlations. This recovers the latent associations underlying the ordered response categories without the attenuation that Pearson coefficients introduce for categorical data. Estimating each correlation on all respondents who answered the relevant item pair retains the full $N = 5{,}153$ module sample. Respondents who skipped or answered "not sure" to an occasional item still contribute to every other association and are assigned factor scores from their observed responses, rather than being removed by listwise deletion. Two design features bound what the analysis can show. First, the 52 items are application- and context-specific by design: the battery contains no multi-item general-attitude instrument, so the analysis characterizes the structure of application-level evaluations and does not evaluate general-attitude scales. Second, the survey administered items in topic blocks. A structure driven by block administration would produce one factor per block; instead, facial recognition alone divides into four factors within a single block, so the recovered structure cuts within blocks rather than tracking them.

Once the factor structure is estimated, we examine the structural relationships among the resulting latent dimensions by estimating a *belief-system network* following the conventions of Epskamp and Fried (2018) and the political-attitude applications of Brandt et al. (2019); Brandt (2020); Turner-Zwinkels and Brandt (2022). Edges in this network are regularized partial correlations. The partial correlation between two factors F_i and F_j is the association that remains between them after the influence of every other factor has been removed, and edges whose magnitude does not exceed sampling noise are shrunk to zero so that only the structurally meaningful associations are retained.[1]

[1]Estimation uses EBICglasso (Epskamp and Fried, 2018) on the oblimin factor-correlation matrix Φ with the conventional EBIC tuning hyperparameter $\gamma = 0.5$ (Foygel and Drton, 2010). Edge-weight and centrality uncertainty are quantified by a parametric bootstrap of $B = 1{,}000$ draws from $\mathrm{MVN}(0, \Phi)$ at the EFA sample size; centrality stability is summarized by the CS-coefficient (Epskamp et al., 2018).

Bivariate correlations between two factors describe how their scores co-vary across respondents, but they do not distinguish a direct relationship between the two factors from one that operates through the rest of the factor system. For instance, two factors representing distinct AI application domains can appear positively correlated simply because both share variance with a third factor that captures a general evaluative tendency. Conversely, two factors that are genuinely related can have their relationship inflated or attenuated by what they share with the remaining factors. The partial-correlation network addresses this by reporting, for each pair of factors, the association that remains once the contribution of every other factor has been accounted for. This is the natural object for characterizing the structure of inter-factor relationships in a multidimensional attitude system. For comparison we report the bivariate factor-correlation graph and an unregularized partial-correlation network in Appendix F; the bivariate graph is included as a visual reference and no centrality or distance metric is computed on it.

Finally, we examine how the latent dimensions are socially distributed. Each of the ten factor scores, z-standardized, is regressed on the same battery of demographic and political predictors: age category, gender, race and ethnicity, educational attainment, family income tier, party identification, ideology, internet use frequency, religious affiliation, religious service attendance, census region, metropolitan status, and marital status (31 terms; Appendix Table H.10). Models are estimated by weighted least squares using the survey weight (`WEIGHT_W99`) with heteroskedasticity-consistent (HC3) standard errors. Respondents are included listwise across predictors, and respondents reporting their gender "in some other way" are excluded because the category is too small to estimate ($n = 31$ in the module sample); each model therefore has $N = 4{,}568$. Throughout, higher scores are pro-AI for F_1–F_4, F_6, F_7, and F_{10}, anti-AI for F_5 and F_9, and F_8 carries no attitudinal valence. These regressions are descriptive: they map which social cleavages align with which attitude dimensions and support no causal interpretation.

3. Results

Our results indicate that AI attitudes cluster not only along the specific technological domains of autonomous vehicles, face recognition, and algorithmic decision-making but also along specific social contexts and areas of application (see Table C.3 in the Appendix for all factor loadings). Take, for example, the set of questions regarding facial recognition technology. Respondents' perception of this technology resolves into four distinct factors, each tied to a different social context of use rather than to the technology as such. While the questions measure respondents' attitudes toward facial recognition, the social context produces four distinct clusters of opinion.

Considering the conventional criteria (Howard, 2016; Ruscio and Roche, 2012), we retained ten factors (see Table A.1 for the adopted factor labels and their highest-loading items, and Table K.16 for the factor-correlation matrix), which explain about 40% of the total variance.

Table C.3 reports the full loading matrix behind these labels. Within this battery of application-specific items, no factor absorbs the shared variance of the others: the ten factors are instead specific to the three substantive domains of the survey (driverless cars, facial recognition, and AI-based algorithms), with F_8 (Self-Reported AI Awareness) entering as a distinct knowledge dimension. Instruments developed to measure a broad orientation toward AI as a general object (Schepman and Rodway, 2020; Sindermann et al., 2021; Stein et al., 2024; Grassini, 2023) target a construct that this battery does not include and that this analysis does not evaluate; multidimensionality has precedent at the instrument level as well, where the AI anxiety scale of Wang and Wang (2022) is itself four-dimensional. Within each domain, multiple correlated yet distinct factors emerge, defined by their social context. Facial recognition

is the clearest case, resolving into four context-specific factors: public-space surveillance (F_4 – Public Surveillance Acceptance), commercial and everyday applications (F_6 – Commercial Facial Recognition Acceptance), police performance (F_7 – Police FaceRec Performance), and anticipated harms (F_9 – Surveillance Harm Anticipation). A recurring distinction is individual versus public utility (e.g., F_1 – Autonomous Vehicle Support and F_3 – Practical AI Optimism); within facial recognition, the same distinction separates accepting public surveillance (F_4) from judging police performance (F_7).

The strongest factor correlations are interpretable in both within- and cross-domain terms. The largest is within the social-media-algorithm domain, between F_2 (Algorithmic Decisions Support) and F_{10} (Content Moderation Efficacy, $r = 0.68$). The next-largest is cross-domain, between F_1 (Autonomous Vehicle Support) and F_6 (Commercial Facial Recognition Acceptance, $r = 0.57$), interpretable as a shared acceptance of everyday and commercial AI deployment. Most other correlations are weak: the factors capture distinct, context-bound attitudes rather than sharing a single dominant source of covariance.

Figure 1 renders these relationships as a regularized partial-correlation network, with edges representing conditional associations between factors after partialing out the other eight. The two strongest conditional edges are F_2–F_{10} ($\rho = 0.59$, 95% bootstrap CI 0.57–0.61), linking algorithmic-decisions support to content-moderation efficacy within the social-media-algorithm domain, and the cross-domain F_1–F_6 ($\rho = 0.48$, CI 0.46–0.50), linking autonomous-vehicle support to commercial facial-recognition acceptance. No single factor acts as a high-strength hub uniting the others (Table F.7); the network has structure, but no factor functions as a general-attitude hub within this ten-factor system. By expected influence the most central nodes are F_{10} (Content Moderation Efficacy) and F_6 (Commercial Facial Recognition Acceptance), whereas F_4 (Public Surveillance Acceptance) and F_5 (Content Filtering Concern) are the most peripheral; by raw node strength the network is instead led by F_1 (Autonomous Vehicle Support) and F_2 (Algorithmic Decisions Support) (Table F.7).

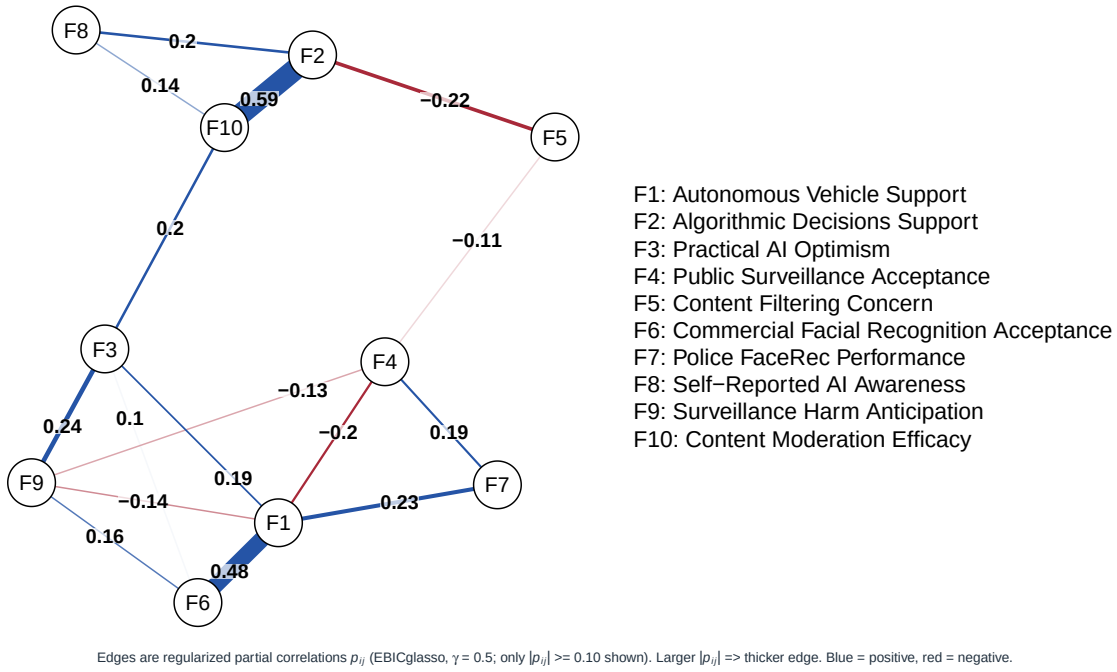


Figure 1: Belief-system network over the ten latent factors. Edges are regularized partial correlations obtained by applying EBICglasso ($\gamma = 0.5$) to the oblimin factor-correlation matrix Φ, following the psychometric-network conventions of Epskamp and Fried (2018) and Brandt et al. (2019). Blue edges denote positive partial correlations, red edges negative; edge thickness scales with $|\rho_{ij}|$. Edge stability and centrality stability are reported in Appendix F.3; the bivariate factor-correlation graph and unregularized partial-correlation network are reported alongside in Appendix F.1 and Appendix F.2 for comparison.

We calculated factor scores for each of the ten attitudes to examine differences among the identified attitudes toward AI. A factor score summarizes how a respondent feels about one specific AI attitude. A factor score for an attitude is determined by a linear combination of the item values and their respective factor loadings on that attitude. This method is similar to a composite scale of the items but also considers the varying degrees of correlation between the items and the specific attitude they map onto. We z-standardized the factor scores for ease of comparison and plot the distributions of the standardized factor scores. These plots are included in Figures G.8 and G.9 in the Appendix. Across the ten factors the distributions differ not only in central tendency but in shape and modality: F_2 (Algorithmic Decisions Support) is sharply right-skewed (skew = 0.96), and both F_2 and F_4 (Public Surveillance Acceptance) exceed the conventional bimodality threshold ($BC = 0.61$ and 0.59, both $> 5/9 \approx 0.556$), the latter the most broadly dispersed and least consensual of the ten (excess kurtosis $= -1.24$, agreement $A = 0.04$). The remaining factors are unimodal, ranging from near-symmetric and broadly dispersed (low agreement) to unimodally concentrated (high agreement), with F_{10} (Content Moderation Efficacy) the next most right-skewed (skew = 0.51).[2]

The regressions test whether the ten dimensions are socially distinguishable. If a single pro/anti disposition organized AI attitudes, any predictor associated with that disposition would move every same-valence factor in the same direction (per the valence convention stated in the study-design section). The coefficients show something else: several predictors carve the attitude space at different angles rather than lining up against a single pro/anti-AI axis.

[2]Table G.9 reports the per-factor skewness, excess kurtosis, bimodality coefficient, normalized entropy, and Van der Eijk agreement statistic.

Age is the cleanest example. Relative to respondents aged 18–29, those aged 50–64 are less optimistic about practical AI applications (F_3, $\beta = -0.46$, $p < .001$) and less convinced that content moderation works (F_{10}, $\beta = -0.39$, $p < .001$), a pattern respondents 65 and older repeat on F_3 (-0.37, $p < .001$) and F_{10} (-0.34, $p < .001$). Yet the same older respondents are *more* accepting of commercial facial recognition (F_6, $\beta_{50\text{–}64} = +0.21$, $p < .05$; $\beta_{65+} = +0.41$, $p < .001$) and rate police facial recognition as more performant (F_7, $\beta_{65+} = +0.30$, $p < .01$). These factors share the same pro-AI valence, yet the age coefficients run in opposite directions, and the cross-equation differences are themselves significant ($p < .001$; Table J.14 in the Appendix). Collapsed into a single general index over the same 52 items, the reversal compresses to one attenuated slope ($\beta_{65+} = -0.19$; Table J.15), understating the movement on both sides.

Ideology (higher = more liberal) spreads across several factors rather than tracking a single axis. More liberal respondents are less concerned about over-filtering (F_5, $\beta = -0.19$, $p < .001$, i.e., more accepting of content moderation) and slightly more supportive of autonomous vehicles (F_1, $\beta = +0.09$, $p = .001$), yet they anticipate *more* surveillance harm (F_9, $\beta = +0.11$, $p < .01$): pro-AI on some attitudes, skeptical on the surveillance-adjacent ones. The raw-sign reversal between the vehicle and commercial facial-recognition slopes does not rest on the modest F_6 coefficient alone ($\beta = -0.09$, $p < .05$); the cross-equation difference between the two ideology slopes is significant at $p < .001$ (Table J.14). Party identification and ideology, entered simultaneously, also structure different subsets of factors: partisanship dominates the content-moderation factors (F_5, F_{10}), while ideology structures the vehicle and surveillance-adjacent dimensions.

The sharpest cross-direction split is racial, and it survives a stricter evidentiary threshold. Black respondents (vs. white) rate police facial recognition as *less* performant (F_7, $\beta = -0.32$, $p < .01$) yet are *more* confident that content moderation works (F_{10}, $\beta = +0.32$, $p < .01$), and the cross-equation difference is significant at $p < .001$. Higher surveillance-harm anticipation (F_9, $\beta = +0.25$, $p < .05$) is consistent with this pattern but rests on a weaker estimate. Institutional skepticism toward policing pairs with openness to content moderation.

Education and gender show the largest and most uniform gaps. Relative to respondents without a high-school degree, every higher attainment level scores lower on commercial facial-recognition acceptance (F_6, β from -0.20 to -0.41, significant from high-school graduates upward) and on content-moderation efficacy (F_{10}, -0.16 to -0.48); because the reference category is small, we read the pattern across levels rather than leaning on any single contrast. Women score half a standard deviation lower than men on autonomous-vehicle support (F_1, $\beta = -0.51$, $p < .001$) and on self-reported AI awareness (F_8, $\beta = -0.52$, $p < .001$). The F_8 gap is a gap in *self-reports*: if men over-report familiarity with AI, part of the other gender differences could reflect response style rather than attitude. The small pro-surveillance difference among women (F_4, $\beta = +0.09$, $p < .05$) does not survive a stricter threshold.

Not every predictor splits this way. Once valence is corrected, Democratic Party identification points in the same pro-AI / pro-moderation direction on most of the factors where it is significant: markedly *lower* F_5 (Content Filtering Concern, $\beta = -0.74$, $p < .001$, the largest coefficient in the analysis), and higher F_{10} (Content Moderation Efficacy, $\beta = +0.33$, $p < .001$), F_1 (Autonomous Vehicle Support, $\beta = +0.20$, $p < .001$), F_2 (Algorithmic Decisions Support, $\beta = +0.19$, $p < .05$), and F_3 (Practical AI Optimism, $\beta = +0.15$, $p < .05$). The one significant exception runs the other way: Democrats anticipate *more* surveillance harm (F_9, $\beta = +0.20$, $p < .01$) – the anti-surveillance direction. Figure 2 renders the full predictor-by-factor grid: each predictor is significant on some factors and null on others, and the predictors that reverse direction pick out different subsets of attitudes from those that do not. Demographic and partisan cleavages therefore do not collapse onto a single pro/anti-AI dimension: each predictor selects

its own subset of factors, and several move factors of the same valence in opposite directions.[3]

The predictors' collective reach is as uneven as any single coefficient. The same battery explains a third of the variance in filtering concern (F_5, adjusted $R^2 = .33$), driven by the party gap that is the largest coefficient in the analysis, but almost none of the variance in public-surveillance acceptance (F_4, $R^2 = .04$), where the party coefficient is null ($\beta = -0.05$, $p = .45$). In November 2021, partisan conflict had organized content-moderation attitudes while leaving facial-recognition attitudes largely unsorted by party or ideology. F_5's predictability is itself party-driven: the factor behaves as a partisan object more than a technology object, which is precisely the kind of distinction a single AI-attitude score cannot represent.

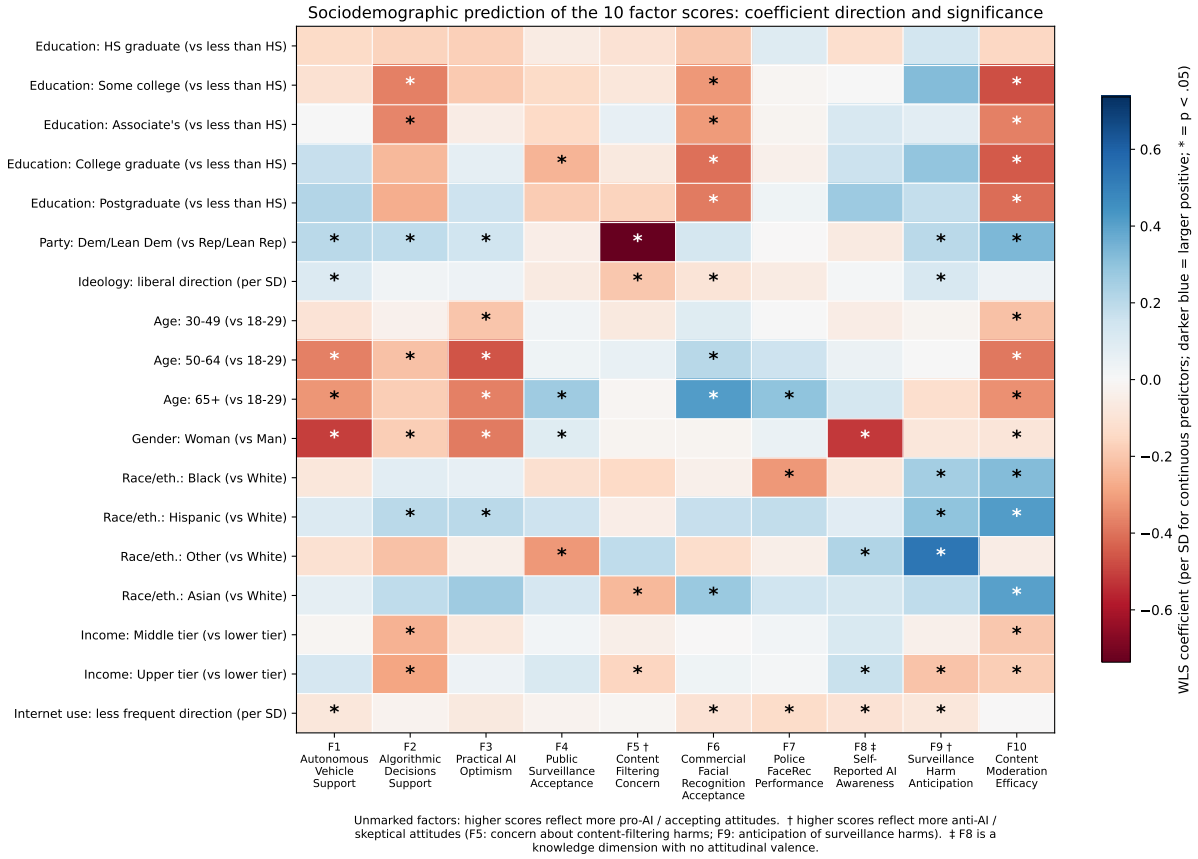


Figure 2: Coefficient grid for the ten factor-score regressions (HC3 standard errors; $N = 4{,}568$ per model). Darker blue encodes larger positive coefficients, darker red larger negative ones; asterisks mark $p < .05$. Continuous predictors are scaled per standard deviation; dummy contrasts are raw, with reference categories in parentheses. The same predictors reverse sign across factors.

4. Discussion

Our analysis maps the structure of US public attitudes toward artificial intelligence in three layers. Fifty-two application-specific survey items resolve into ten factors that separate by the social context in which a technology is deployed rather than by the technology itself. The

[3]Across the ten models, 77 of 310 coefficients are significant at $p < .05$, against roughly 16 expected by chance under a global null. Headline claims in the text are held to $p < .01$ or to the formal cross-equation tests in Table J.14. Full coefficient tables appear in Tables I.11 and I.12.

partial-correlation network among these factors has clear structure, but no factor functions as a general-attitude hub within this ten-factor system. And the social cleavages that organize these attitudes cut across the factors: the same demographic and political groups hold accepting positions on some dimensions and skeptical positions on others. An influential line of work has developed instruments that summarize attitudes toward AI as a general disposition (Schepman and Rodway, 2020; Stein et al., 2024; Grassini, 2023; Sindermann et al., 2021). Such instruments are well suited to tracking overall sentiment and comparing populations. What they are not designed to settle is whether attitudes hold together across AI domains and deployment contexts. That is the question we take up here, and our answer complements general measurement rather than replacing it.

The three layers organize the evidence as follows:

(a) *Constituent attitudes and where they cut.* The ten factors partition by context of use, not by technology. Facial recognition alone divides into four dimensions: acceptance of public-space surveillance (F_4), acceptance of commercial and everyday uses (F_6), judgments of whether the technology performs accurately for police (F_7), and anticipation of surveillance harms (F_9). Content moderation divides into concern about over-filtering (F_5) and judgments of efficacy (F_{10}). Self-reported AI awareness (F_8) is a distinct knowledge dimension with no attitudinal valence.

(b) *Conditional connectivity and the absent hub.* Once every other factor is held constant, the strongest conditional associations connect substantively adjacent judgments: support for algorithmic decisions with content-moderation efficacy ($\rho = .59$ between F_2 and F_{10}), and autonomous-vehicle support with commercial facial-recognition acceptance ($\rho = .48$ between F_1 and F_6). No factor's removal would disconnect the system, and none functions as a hub through which the others relate.

(c) *Social distribution and cross-cutting cleavages.* Age, education, partisanship, gender, and race each structure some dimensions strongly and others not at all, and several move dimensions of the same pro-AI valence in opposite directions.

The domains differ not only in which concerns they engage but in how tightly those concerns cohere, and this asymmetry is itself informative. Driverless vehicles are the least fractionated domain: ten of the thirteen attitudinal vehicle items load on a single support factor (F_1) between .61 and .90. The three exceptions are diagnostic, since the items on hacking risk, job loss, and income inequality all lean toward general anticipation of surveillance harms (F_9). Facial recognition, by contrast, fractures into the four dimensions above, and content moderation separates concern from efficacy, with the concern factor anchored by the items measuring censorship worry (.77) and wrongful-removal worry (.79). Which considerations cohere also crosses domain boundaries: the second-strongest conditional association in the network links vehicle support to commercial facial recognition, so domain boundaries organize the factors without sealing them. One measured belief deserves separate mention: the item asking whether police would use facial recognition to monitor Black and Hispanic neighborhoods much more often than other neighborhoods loads on harm anticipation (.63 on F_9), and Black respondents score lower on police facial-recognition performance ($\beta = -0.32$ on F_7) while scoring higher on moderation efficacy (+0.32 on F_{10}), a pattern consistent with skepticism directed at policing institutions specifically rather than at technology in general. These are covariation patterns in one cross-section; claims about the processes that generate them require experimental or panel data. The survey also fielded no items on privacy as such, so privacy concerns enter this map only through their surveillance-adjacent neighbors.

Read as a map of the considerations each controversy engages, the loadings show three different patterns. For driverless vehicles, safety, usefulness, and personal trust form a single judgment: expected traffic deaths and injuries (.80), willingness to ride (.84), comfort sharing the road (.81), and greater independence for older and disabled people (.61) all load on the same support factor (F_1). Two concerns sit apart from that judgment: whether the vehicles could be hacked in ways that put safety at risk, which loads partly on harm anticipation (.38 on F_9), and whether drivers would lose their jobs, which loads mainly there (.34). Facial recognition separates crime control from surveillance. Expectations that police would solve crimes faster (.67) and find more missing persons (.60), together with the expected effect on crime (.44), define the performance factor (F_7), while expectations that police would track everyone's location (.53), make more false arrests (.58), and monitor Black and Hispanic neighborhoods more often (.63) define harm anticipation (F_9); whether policing would become fairer loads on both (.40 and −.37). Content moderation separates censorship from information quality. Worry that political viewpoints are censored (.77) and that information is wrongly removed (.79) defines filtering concern (F_5), while judgments that trustworthy information has become easier to find (.53) and that conversations have become more meaningful (.56) define efficacy (F_{10}). Partisan conflict attaches to the censorship side: the party gap on F_5 is the largest coefficient in the analysis. Beyond privacy, the battery has no items on personal autonomy as such or on trust in the platform companies, so those concerns cannot be located here. For measurement, prediction, and governance alike, the consequence is that each controversy engages its own considerations, and an instrument, forecast, or consultation framed around "AI" in general cannot show which consideration organizes a given dispute.

No demographic or political group in this sample is uniformly for or against AI. Partisanship saturates content moderation, where the Democratic-Republican gap on filtering concern is the largest coefficient in the analysis ($\beta = -0.74$ on F_5, model adjusted $R^2 = .33$), while leaving public-surveillance acceptance essentially unstructured ($\beta = -0.05$, n.s.; $R^2 = .04$): as of late 2021, facial-recognition attitudes remained available for polarization rather than already sorted by party. Older Americans are more skeptical specifically of novel algorithmic applications, not of AI as such: respondents 65 and older score lower on practical AI optimism ($\beta = -0.37$ on F_3) yet higher on commercial facial-recognition acceptance (+0.41 on F_6), and the cross-equation difference is itself significant ($p < .001$; Appendix Table J.14). The measurement implication follows directly, and we can state it empirically rather than hypothetically: when the same 52 items are collapsed into a single general index (the first principal component) and regressed on the same predictors, the ideology coefficient collapses to null (+0.05, n.s.), the Black-White difference vanishes (−0.04, n.s.), and the education gradient flattens to non-significance. Gender, age, and partisanship survive the collapse, as do Hispanic identity, Protestant affiliation, and internet use (Appendix Table J.15). A one-dimensional reading of these data would report no ideological structure, no Black-White difference, and no educational pattern. All three are present, large, and interpretable at the factor level. For consultation exercises built around "the public's view of AI," the practical consequence is that the constituencies who would support or oppose a specific deployment are identifiable, and they differ from one application to the next.

Three implications follow, each anchored in a specific result. First, for measurement: general and application-specific items are not interchangeable. The survey's own global item, which asks whether AI makes respondents more excited or more concerned, shares almost no variance with the rest of the battery (communality = .05), while the specific items resolve into ten interpretable factors. Instruments designed for application-level questions should stratify by deployment context, and should measure acceptance separately from efficacy: within policing alone, whether facial recognition works (F_7) and whether it should be used in public spaces

(F_4) are distinct dimensions. We also report a negative finding of use to instrument designers: the items intended to tap abstract algorithmic fairness and human-versus-machine decision preference failed to cohere with any factor (communalities .15 and .18), so constructs of that kind appear to require purpose-built batteries rather than single items. The ten factors reported here are one sample's realization of a design principle (organization by context of use), and the principle, not the particular solution, is what we expect to generalize; confirmation awaits factor analysis in an independent sample.

Second, for the study of opinion structure: demographic and political position locates people well on some AI attitudes and almost not at all on others. The same 31-predictor battery explains a third of the variance in filtering concern ($R^2 = .33$) and almost none of the variance in surveillance acceptance ($R^2 = .04$). These are cross-sectional associations estimated without out-of-sample validation, so we describe attitudes as socially structured, not as predictable from demographics. Within this battery, the pattern rules out an account on which a single pro-anti disposition, socially located in one set of groups, organizes application-level evaluations.

Third, for governance: "public support for AI" is not a well-defined input to policy, because the level of consensus is itself application-specific. Within the same policing technology, the public is close to consensus that facial recognition performs effectively for police (Van der Eijk agreement $A = .57$ for F_7) while being almost evenly divided over whether public-space scanning is acceptable ($A = .04$, with the bimodality coefficient exceeding the conventional threshold, for F_4). A single headline support number for such a technology is an artifact of which question was asked. Democratic accountability in this domain begins with opinion inputs measured at the resolution at which the public actually disagrees — the deployment context — and the factor structure reported here indicates what that resolution is.

Several limitations bound these conclusions. The data are a single cross-section from November 2021, and the survey did not cover job displacement, military applications, AI ethics debates, artificial general intelligence, or privacy as a named construct. The network's edges are conditional associations, not causal pathways, and the factor solution is exploratory and rotation-contingent: an oblique rotation absorbs shared variance into factor correlations by construction, so this design characterizes structure within an application-specific battery and cannot rule out a superordinate disposition that such a battery does not measure. A second-order analysis of the factor correlations (Appendix Table J.13) indicates that any general factor above the ten is weak, accounting for under a fifth of the factor-level variance, but the definitive test is a bifactor or higher-order model fit to new data. The specific ten-factor map is indexed to this item pool and this moment; what we expect to travel are the principles of context-of-use organization, the absence of a hub within application-level batteries, and cross-cutting cleavages. Finally, these data predate the mass adoption of large language models. A general AI attitude may have consolidated since, precisely because a shared public referent for "AI" now exists; whether the structure documented here has tightened, loosened, or reorganized is a question for the next wave of data, as is whether the network's connectivity varies with political sophistication or AI awareness.

5. Conclusion

We asked four questions about American attitudes toward artificial intelligence, namely whether they rest on a single evaluative dimension or divide by domain, whether the social context of an application matters within a domain, how the resulting attitudes connect, and whether the social and political cleavages around AI line up along one pro-anti axis. The four answers are consistent with one another within the Pew battery fielded in November 2021.

Attitudes divide by domain and, within a domain, by the context in which an application is used. The resulting dimensions connect to one another without a hub. The groups that accept one application often resist another. Taken together, these answers indicate that opinion on AI in this period resembles a set of positions on distinct policy questions more than a single attitude toward a single object.

These findings complement the general-attitude scales widely used to measure AI opinion. These scales record whether people hold a broad orientation toward AI. That quantity is well suited to tracking sentiment over time and across populations. Our results concern a different level, the one at which citizens evaluate particular deployments. Each level corresponds to a different question. AI opinion in 2021 was weakly constrained across applications, in the sense of the belief-system tradition, as Converse (1964) found for mass political attitudes. It was nonetheless structured, in that attitudes cohered within contexts of use and connected across them through a few substantive links.

Because opinion is structured this way, a poll that asks whether Americans support AI returns a figure that reflects whichever applications come to respondents' minds. In these data, the public broadly agreed that police facial recognition works but was nearly evenly divided over scanning people in public places. Additionally, older Americans, who doubted algorithmic applications, were more accepting of commercial facial recognition than younger Americans.

A single support figure averages over these differences. Hence, anyone deciding whether to deploy, regulate, or contest a particular application needs to know how the public evaluates that application in the setting of its use.

This structure of opinion has three political implications. First, AI does not divide this public into a pro-technology camp and an anti-technology camp. Alignments form application by application. Compared with white respondents, for example, Black respondents were more skeptical of police facial recognition and more confident in content moderation, whereas concern about content filtering was organized above all by party. Under such cross-cutting alignments, few citizens are opposed across the board. This pattern can reduce conflict over AI as a whole even as disputes over particular uses grow sharper.

Second, partisanship had organized the controversy over content moderation but not yet the controversy over the use of facial recognition in public spaces. Where party cues have not attached, opinion remains available for polarization. The partisan alignment of later AI controversies is therefore still open. A central question for the politics of AI is which applications become partisan objects, and when.

Third, democratic accountability for AI depends on consulting the public deployment by deployment. Rules that treat AI as a single object would answer to an average opinion held by few constituencies. Rules written for specific uses instead match the level at which citizens form their views and disagree about them, as in the case of police surveillance, commercial identification, and automated decisions about credit, medical treatment, hiring, and parole.

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Appendix A. Factor-Labeling Rationale

Table A.1: Factor-labeling rationale: adopted labels and the three highest-loading items per factor. Item loadings are from the oblimin-rotated 10-factor solution.

Factor	Adopted Label	Item Wording	Loading
F_1	Autonomous Vehicle Support	• Do you think widespread use of driverless passenger vehicles would be a good idea or bad idea for society?	0.904
		• Would you personally want to ride in a driverless passenger vehicle, if you had the opportunity?	0.837
		• If the use of driverless passenger vehicles became widespread, how comfortable would you feel sharing the road with them?	0.813
F_2	Algorithmic Decisions Support	• Would you favor or oppose the use of computer programs to make final decisions about which people should be good candidates for parole?	0.842
		• Would you favor or oppose the use of computer programs to make final decisions about which job applicants should move on to a next round of interviews?	0.828
		• Would you favor or oppose the use of computer programs to make final decisions about which people should be approved for mortgages?	0.817
F_3	Practical AI Optimism	• How excited or concerned would you be if AI computer programs could perform household chores?	0.728
		• How excited or concerned would you be if AI computer programs could perform repetitive workplace tasks?	0.728
		• How excited or concerned would you be if AI computer programs could make important life decisions for people?	0.654
F_4	Public Surveillance Acceptance	• Would you consider the use of facial recognition technology by police to be acceptable or not acceptable? Scanning people as they enter large events like concerts to see who is in the crowd	0.940
		• Would you consider the use of facial recognition technology by police to be acceptable or not acceptable? Scanning people at public protests	0.870
		• Would you consider the use of facial recognition technology by police to be acceptable or not acceptable? Scanning people as they walk down the street	0.780
F_5	Content Filtering Concern	• Do you think widespread use of computer programs by social media companies to find false information is making the following happen? News and information are being wrongly removed	0.793
		• Do you think widespread use of computer programs by social media companies to find false information is making the following happen? Political viewpoints are being censored	0.767
		• Do you think widespread use of computer programs by social media companies to find false information on their sites has been a good idea or bad idea for society?	−0.671
F_6	Commercial Facial Recognition Acceptance	• Would you favor or oppose the use of facial recognition technology for each of the following purposes? Companies automatically tracking the attendance of their employees	0.736
		• Would you favor or oppose the use of facial recognition technology for each of the following purposes? Apartment buildings tracking who enters or leaves their buildings	0.720
		• Would you favor or oppose the use of facial recognition technology for each of the following purposes? Retail stores enhancing credit card payment security by confirming account holders at checkout	0.708
F_7	Police FaceRec Performance	• If the use of facial recognition technology by police becomes widespread, do you think the police would solve crimes more quickly and efficiently?	0.666
		• If the use of facial recognition technology by police becomes widespread, do you think the police would find more missing persons?	0.601
		• If the use of facial recognition technology by police becomes widespread, do you think crime would increase in the U.S., decrease in the U.S., or stay about the same?	0.442
F_8	Self-Reported AI Awareness	• How much have you heard or read about facial recognition technology by police?	0.804
		• How much have you heard or read about driverless passenger vehicles?	0.738
		• How much have you heard or read about computer programs used by social media companies to find false information on their sites?	0.670
F_9	Surveillance Harm Anticipation	• If the use of facial recognition technology by police becomes widespread, do you think the police would use the technology to monitor Black and Hispanic neighborhoods much more often than other neighborhoods?	0.627
		• If the use of facial recognition technology by police becomes widespread, do you think the police would make more false arrests?	0.582
		• If the use of facial recognition technology by police becomes widespread, do you think the police would be able to track everyone's location at all times?	0.527

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Table A.1 continued

Factor	Adopted Label	Item Wording	Loading
F_{10}	Content Moderation Efficacy	• Do you think widespread use of computer programs by social media companies to find false information is making the following happen? It is allowing people to have more meaningful conversations	0.560
		• Do you think widespread use of computer programs by social media companies to find false information is making the following happen? It is getting easier to find trustworthy information	0.530
		• How excited or concerned would you be if AI computer programs could know people's thoughts and behaviors?	0.328

Note. F_1: Autonomous Vehicle Support; F_2: Algorithmic Decisions Support; F_3: Practical AI Optimism; F_4: Public Surveillance Acceptance; F_5: Content Filtering Concern; F_6: Commercial Facial Recognition Acceptance; F_7: Police FaceRec Performance; F_8: Self-Reported AI Awareness; F_9: Surveillance Harm Anticipation; F_{10}: Content Moderation Efficacy. Higher scores reflect more pro-AI / accepting attitudes for F_1, F_2, F_3, F_4, F_6, F_7, F_{10}, and more anti-AI / skeptical attitudes for F_5 (concern about content-filtering harms) and F_9 (anticipation of surveillance harms). F_8 is a knowledge dimension with no attitudinal valence.

Appendix B. Survey Items

Table B.2: Complete inventory of survey items included in the exploratory factor analysis.

Item	Item Wording	Type	Rev.
General AI			
ai_cap_customer_calls	How excited or concerned would you be if AI computer programs could *handle customer service calls?*[a]	Attitude	No
ai_cap_diag_medical	... *diagnose medical problems?*[a]	Attitude	No
ai_cap_house_chores	... *perform household chores?*[a]	Attitude	No
ai_cap_important_decis	... *make important life decisions for people?*[a]	Attitude	No
ai_cap_know_thoughts_behav	... *know people's thoughts and behaviors?*[a]	Attitude	Yes
ai_cap_repeti_work_task	... *perform repetitive workplace tasks?*[a]	Attitude	No
ai_excitement_concern	Overall, would you say the increased use of artificial intelligence computer programs in daily life makes you feel more excited than concerned, more concerned than excited, or equally concerned and excited?[b]	Attitude	No
algor_fair	Do you think it is possible or not possible for people to design artificial intelligence computer programs that can consistently make fair decisions in complex situations?[c]	Belief	No
Social Media Algorithms			
ai_pref_medicaldecis	Would you favor or oppose the use of computer programs to make final decisions about *which patients should get a medical treatment?*[d]	Attitude	No
ai_pref_mortgages	... *which people should be approved for mortgages?*[d]	Attitude	No
ai_pref_parole	... *which people should be good candidates for parole?*[d]	Attitude	No
ai_pref_recruiting	... *which job applicants should move on to a next round of interviews?*[d]	Attitude	No
aism_idea_judgment	Do you think widespread use of computer programs by social media companies to find false information on their sites has been a good idea or bad idea for society?[e]	Attitude	No
aism_pref_decision	Do you think the decisions made by social media companies about what information is false should be mostly made by people, mostly made by computer programs, or a mix of both?[f]	Attitude	No
aism_algo_accur	When it comes to finding false information on their sites, do you think computer programs used by social media companies do a better job than humans, a worse job, or about the same job?[g]	Belief	No
aism_eval_betterconver	Do you think widespread use of computer programs by social media companies to find false information is making the following happen? *It is allowing people to have more meaningful conversations*[h]	Belief	No
aism_eval_censorship	... *Political viewpoints are being censored*[h]	Belief	Yes
aism_eval_infoquality	... *It is getting easier to find trustworthy information*[h]	Belief	No
aism_eval_mistakes	... *News and information are being wrongly removed*[h]	Belief	Yes
aism_knowledge	How much have you heard or read about computer programs used by social media companies to find false information on their sites?[i]	Knowledge	No

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Table B.2 continued

Item	Item Wording	Type	Rev.
Facial Recognition			
aiface_confidence_arrests	If a facial recognition program said that someone was involved in a crime, should that be good enough evidence for police to arrest them, even if there was a small chance the program was wrong?[j]	Attitude	No
aiface_eval_crowded_events	Would you consider the use of facial recognition technology by police to be acceptable or not acceptable? *Scanning people as they enter large events like concerts to see who is in the crowd*[k]	Attitude	No
aiface_eval_public_protest	. . . *Scanning people at public protests*[k]	Attitude	No
aiface_eval_street	. . . *Scanning people as they walk down the street*[k]	Attitude	No
aiface_idea_judgment	Do you think the widespread use of facial recognition technology by police would be a good idea or bad idea for society?[e]	Attitude	No
aiface_pref_buildingsecurity	Would you favor or oppose the use of facial recognition technology for each of the following purposes? *Apartment buildings tracking who enters or leaves their buildings*[d]	Attitude	No
aiface_pref_creditsecurity	. . . *Retail stores enhancing credit card payment security by confirming account holders at checkout*[d]	Attitude	No
aiface_pref_smphototag	. . . *Social media sites automatically identifying people in photos*[d]	Attitude	No
aiface_pref_workattend	. . . *Companies automatically tracking the attendance of their employees*[d]	Attitude	No
aiface_crimeimpact	If the use of facial recognition technology by police becomes widespread, do you think crime would increase in the U.S., decrease in the U.S., or stay about the same?[l]	Belief	No
aiface_eval_crowdcontrol	If the use of facial recognition technology by police becomes widespread, do you think the police would *be better able to keep crowds under control?*[m]	Belief	No
aiface_eval_efficiency	. . . *solve crimes more quickly and efficiently?*[m]	Belief	No
aiface_eval_farrests	. . . *make more false arrests?*[m]	Belief	Yes
aiface_eval_missingpers	. . . *find more missing persons?*[m]	Belief	No
aiface_eval_racebias	. . . *use the technology to monitor Black and Hispanic neighborhoods much more often than other neighborhoods?*[m]	Belief	Yes
aiface_eval_tracklocation	. . . *be able to track everyone's location at all times?*[m]	Belief	Yes
aiface_fairness	Do you think the widespread use of facial recognition technology by police will make policing more fair, less fair, or not make much difference?[n]	Belief	No
aiface_knowledge	How much have you heard or read about facial recognition technology by police?[i]	Knowledge	No
Driverless Vehicles			
dcars_idea_judgment	Do you think widespread use of driverless passenger vehicles would be a good idea or bad idea for society?[e]	Attitude	No
dcars_pref_18wheeltruck	Would you favor or oppose the use of driverless vehicle technology for each of the following purposes? *18-wheeler trucks*[d]	Attitude	No
dcars_pref_delivery	. . . *Delivery vehicles*[d]	Attitude	No
dcars_pref_publictransbus	. . . *Buses for public transportation*[d]	Attitude	No
dcars_pref_taxisuber	. . . *Taxis and ride-sharing vehicles*[d]	Attitude	No
dcars_wides_comfortable	If the use of driverless passenger vehicles became widespread, how comfortable would you feel sharing the road with them?[o]	Attitude	No
dcars_would_ride	Would you personally want to ride in a driverless passenger vehicle, if you had the opportunity?[p]	Attitude	No
dcars_deathsimpact	If the use of driverless passenger vehicles becomes widespread, do you think that would increase the number of people killed or injured in traffic accidents, decrease it, or not make much difference?[q]	Belief	No
dcars_eval_commutingstress	If the use of driverless passenger vehicles becomes widespread, do you think the following would happen? *Getting from place to place would be less stressful*[m]	Belief	No
dcars_eval_hackrisk	. . . *The computer systems in driverless passenger vehicles would be easily hacked in ways that put safety at risk*[m]	Belief	Yes
dcars_eval_jobloss	. . . *Many people who make their living by driving others or delivering things with passenger vehicles would lose their jobs*[m]	Belief	Yes
dcars_eval_seniorsindep	. . . *Older adults and people with disabilities will be able to live more independently*[m]	Belief	No

dcars_ inequalityimpact	If the use of driverless passenger vehicles becomes widespread, do you think that would increase the gap between higher and lower-income Americans, decrease it, or not make much difference?[r]	Belief	No
dcars_knowledge	How much have you heard or read about driverless passenger vehicles?[i]	Knowledge	No

Note. Response scales: [a] Very excited / Somewhat excited / Equal / Somewhat concerned / Very concerned; [b] More excited / More concerned / Equally concerned and excited; [c] Possible / Not possible / Not sure; [d] Favor / Oppose / Not sure; [e] Good idea / Bad idea / Not sure; [f] Mostly by people / Mostly by computer programs / A mix of both / Not sure; [g] Better job than humans / Worse job / About the same / Not sure; [h] Definitely happening / Probably happening / Probably NOT happening / Definitely NOT happening; [i] A lot / A little / Nothing at all; [j] Yes / No; [k] Acceptable / Not acceptable; [l] Increase / Decrease / Stay about the same; [m] Definitely would happen / Probably would happen / Probably would NOT happen / Definitely would NOT happen; [n] More fair / Less fair / Not make much difference; [o] Extremely comfortable / Very comfortable / Somewhat comfortable / Not too comfortable / Not comfortable at all; [p] Definitely want / Probably want / Probably NOT want / Definitely NOT want; [q] Increase / Decrease / Not make much difference; [r] Increase the gap / Decrease the gap / Not make much difference. Items are grouped by the questionnaire module in which Pew fielded them. Item names are the analysis dataset's variable identifiers. Three are shown with corrected spelling; their dataset columns are `ai_excitment_concern`, `ai_pref_recruting`, `dcars_pref_18weeltruck`.

Appendix C. Factor Loadings

Table C.3: Oblimin pattern-loading matrix for the 10-factor solution. Loadings with $|\lambda| < 0.15$ are blanked for readability. h^2 is the communality of each item (variance accounted for by the ten retained factors).

Item	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	h^2
ai_excitement_concern											0.05
algor_fair			0.27							0.18	0.15
ai_cap_know_thoughts_behav			0.58							0.33	0.48
ai_cap_house_chores			0.73							−0.18	0.62
ai_cap_important_decis		0.18	0.65							0.32	0.59
ai_cap_diag_medical			0.61								0.40
ai_cap_repeti_work_task			0.73							−0.17	0.59
ai_cap_customer_calls			0.64								0.43
aism_knowledge								0.67			0.46
aism_idea_judgment					−0.67						0.50
aism_eval_mistakes					0.79						0.64
aism_eval_censorship					0.77						0.60
aism_eval_infoquality					−0.25		0.16			0.53	0.39
aism_eval_betterconver					−0.22		0.15			0.56	0.42
aism_pref_decision		−0.21			0.28						0.18
aism_algo_accur		0.23			−0.38		0.22				0.28
ai_pref_mortgages		0.82									0.68
ai_pref_medicaldecis		0.80									0.66
ai_pref_recruiting		0.83									0.69
ai_pref_parole		0.84									0.72
aiface_knowledge								0.80			0.65
aiface_idea_judgment				0.27		0.29	0.37		−0.26		0.38
aiface_eval_farrests							−0.27		0.58		0.43
aiface_eval_efficiency							0.67				0.50
aiface_eval_racebias					−0.21				0.63		0.45
aiface_eval_missingpers							0.60				0.40
aiface_eval_tracklocation					0.21		0.16		0.53		0.37
aiface_eval_crowdcontrol							0.41				0.24
aiface_fairness				0.20		0.17	0.40		−0.37		0.37
aiface_crimeimpact							0.44		−0.20		0.27
aiface_confidence_arrests				0.29					−0.20		0.17
aiface_eval_public_protest				0.87							0.77
aiface_eval_crowded_events				0.94							0.90
aiface_eval_street				0.78							0.63
aiface_pref_workattend						0.74					0.56
aiface_pref_smphototag						0.57				0.16	0.37
aiface_pref_creditsecurity						0.71					0.52
aiface_pref_buildingsecurity						0.72					0.53
dcars_knowledge								0.74			0.55
dcars_idea_judgment	0.90										0.82
dcars_would_ride	0.84										0.72
dcars_eval_seniorsindep	0.61						0.27				0.47
dcars_eval_jobloss							0.25		0.34		0.23
dcars_eval_commutingstress	0.73						0.18				0.59

Continued on next page

Table C.3 continued

Item	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}	h^2
dcars_eval_hackrisk	−0.44								0.38		0.40
dcars_inequalityimpact	−0.21								0.16		0.11
dcars_deathsimpact	0.80										0.67
dcars_wides_comfortable	0.81										0.68
dcars_pref_taxisuber	0.81										0.69
dcars_pref_18wheeltruck	0.78										0.65
dcars_pref_publictransbus	0.76										0.61
dcars_pref_delivery	0.71										0.55

Under oblique rotation, factors are permitted to correlate, and a single item may load above threshold on more than one factor. When such cross-loaders relate positively to one factor and negatively to another, the signs reflect the item's directional relationship with each latent construct—not a coding inconsistency. For example, an item about technology risk may load positively on a harm-anticipation factor and negatively on an adoption-enthusiasm factor, because the same belief (that the technology is risky) runs in opposite directions relative to the two constructs. All but four items load positively on their primary (strongest-loading) factor; the exceptions are risk- or accuracy-worded items whose strongest loading is negative (two driverless-car risk items on F_1 and two social-media-algorithm items on F_5), reflecting the same directional logic described above.

Appendix D. Parallel Analysis

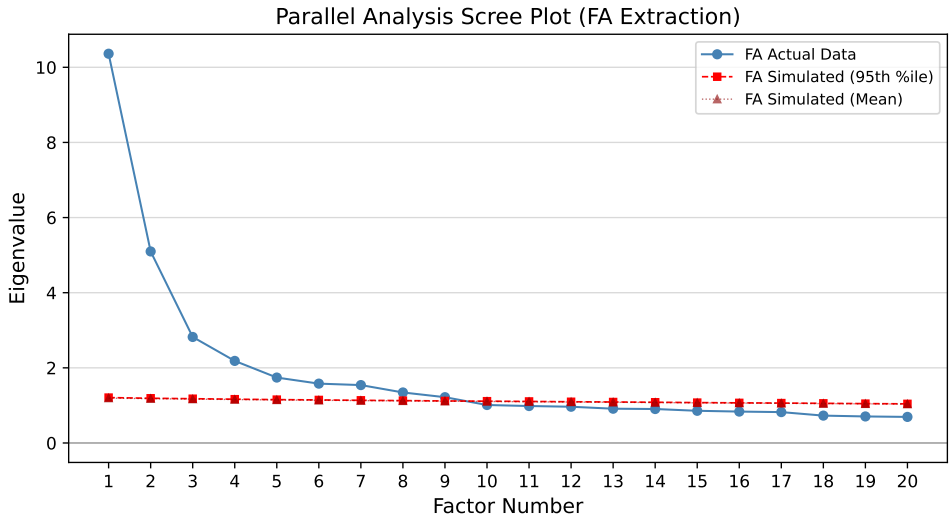


Figure D.3: Parallel analysis scree plot (factor analysis variant). Observed eigenvalues are compared against simulated random data eigenvalues to determine the number of factors to retain.

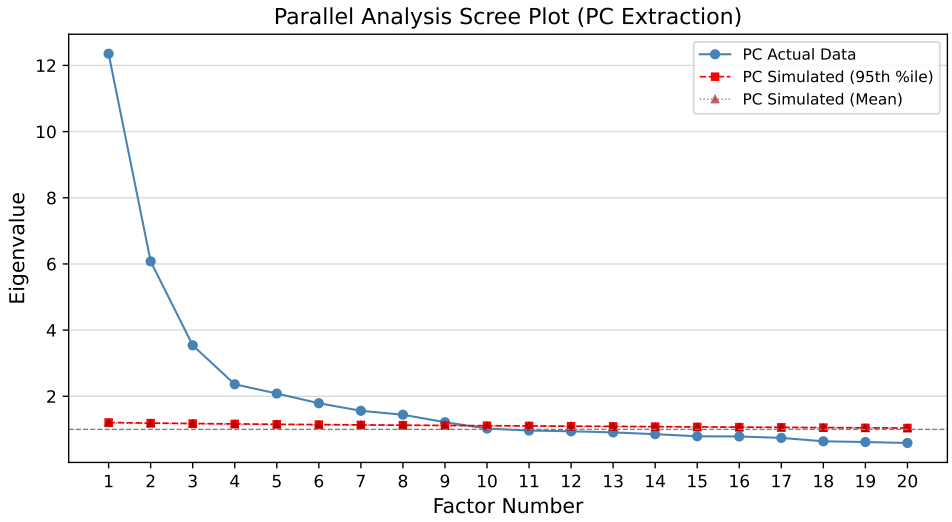


Figure D.4: Parallel analysis scree plot (principal components variant).

Appendix E. 10-Factor Solution Retention Rationale

We used parallel analysis as the primary statistical guide for factor retention because it asks whether each observed eigenvalue is larger than would be expected from random data with the same dimensionality. This criterion is better suited to the present item battery than either an elbow rule or a Kaiser eigenvalue-greater-than-one rule. The elbow rule depends on a visual judgment about where the scree plot begins to flatten, which is difficult to make reproducibly when the eigenvalue profile declines gradually. The Kaiser rule is also too mechanical for this setting: with a large battery of ordinal survey items, it can retain factors that are statistically extractable but substantively thin. Parallel analysis therefore provides the most defensible starting point, while still leaving room for substantive inspection of the rotated solution (Howard, 2016; Ruscio and Roche, 2012).

Parallel analysis pointed to a nine-factor solution, but the rotated nine-factor loadings showed one important compression. The decision to move from nine to ten factors did not simply isolate weak residual items; rather, the nine-factor solution collapses facial-recognition acceptance across very different social contexts onto a single factor. Items about accepting facial recognition in public space—at protests, in crowds, and on the street—load together with items about accepting it for everyday commercial purposes—workplace attendance tracking, apartment-building access, in-store payment security, and social-media photo-tagging (all loading 0.53–0.82 on the combined nine-factor dimension). These are not interchangeable indicators of a single latent attitude: the first concerns state and public surveillance, while the second concerns routine commercial and personal applications, and the ten-factor solution separates them into Public Surveillance Acceptance and Commercial Facial Recognition Acceptance.

For that reason, we retained the 10-factor oblimin solution. The tenth factor matters substantively because it resolves the facial-recognition domain into four context-specific dimensions rather than three: public-space surveillance (F_4 – Public Surveillance Acceptance), commercial and everyday applications (F_6 – Commercial Facial Recognition Acceptance), police performance (F_7 – Police FaceRec Performance), and anticipated harms (F_9 – Surveillance Harm Anticipation). Collapsing to fewer factors merges the public-surveillance and commercial-application items into a single "facial monitoring" factor and obscures one of the central findings of the paper: attitudes toward AI are organized by social context as well as by technology domain and evaluative stance. The 10-factor solution also avoids the over-extraction problem

visible in the 12-factor comparison, where several additional factors are thin and difficult to defend as standalone constructs. We therefore treat the 10-factor solution as a theoretically disciplined refinement of the parallel-analysis recommendation, not as a purely mechanical eigenvalue decision.

Appendix F. Factor-Correlation Diagnostics

We report three views of the structural relationships among the ten latent factors, in order of methodological strength: a bivariate factor-correlation graph (marginal associations), an unregularized partial-correlation network (conditional associations), and the regularized belief-system network estimated in the main text (Figure 1). Each step shrinks the set of substantively meaningful edges by conditioning on more of the other factors. Network-structure metrics (centrality, pairwise distance, edge stability) are computed only on the two conditional graphs; the bivariate graph is presented for comparison.

Appendix F.1. Bivariate Factor-Correlation Graph (Visualization Only)

Figure F.5 re-expresses the oblimin factor-correlation matrix Φ as a thresholded graph retaining edges with $|r_{ij}| \geq 0.10$. Each retained edge is a *marginal* association between two factors and does not distinguish a direct link from one mediated through other factors. We therefore compute no centrality, no pairwise distance, and no path-based metrics on this graph: a high-degree node here may simply be a factor that shares variance with many others through one or two well-connected intermediaries, and a $1/|r_{ij}|$ path distance summed along multi-hop chains would conflate direct and indirect association. Structural metrics live exclusively on the partial-correlation graph (Appendix F.2) and the belief-system network (Appendix F.3).

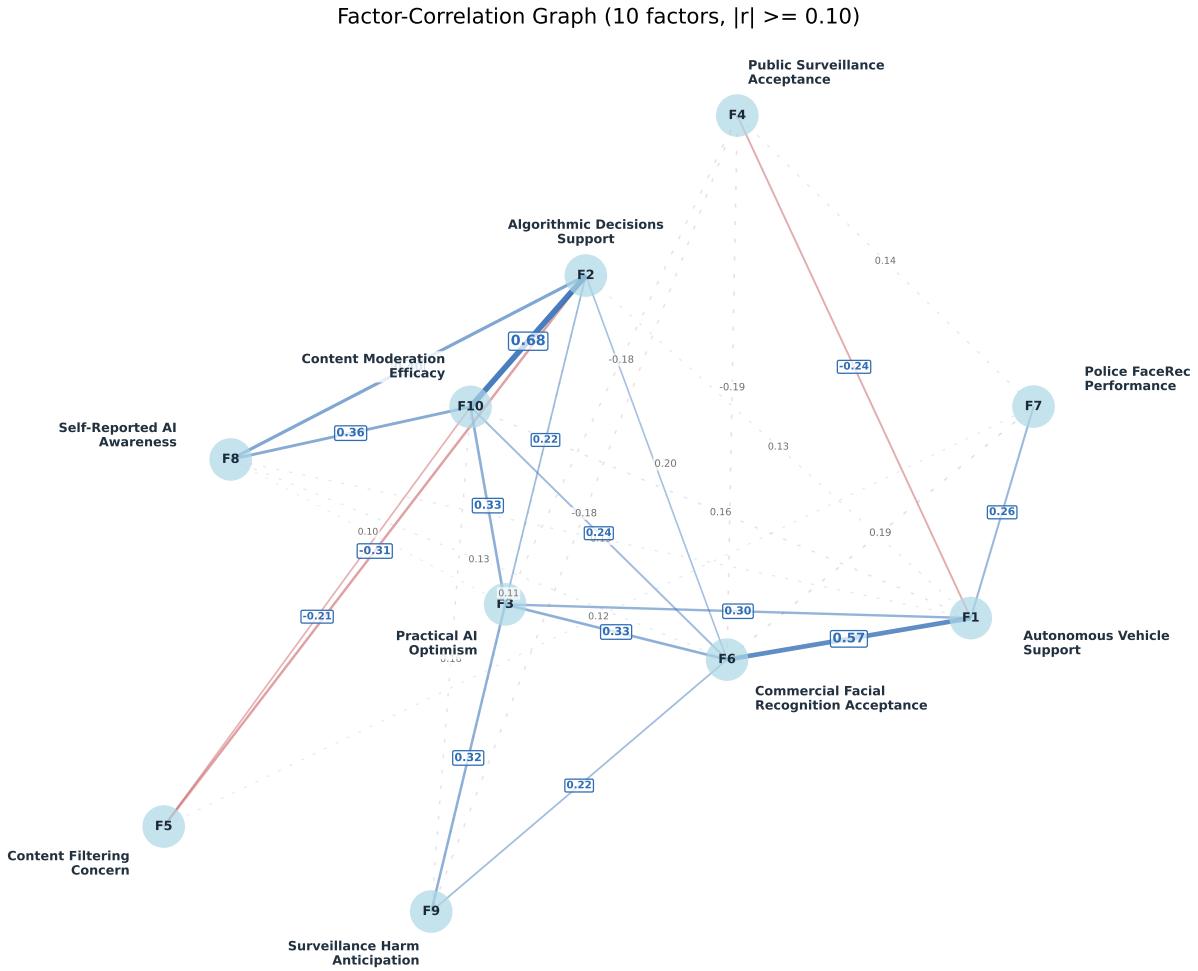


Figure F.5: Ten-factor bivariate factor-correlation graph at $|r_{ij}| \geq 0.10$. Blue edges are positive, red edges negative; edge thickness scales with $|r_{ij}|$. Visualization only: no centrality or path-based metrics are computed on this graph because bivariate r confounds direct and indirect association.

Appendix F.2. Unregularized Partial-Correlation Network

Let $K = \Phi^{-1}$ denote the precision matrix of the latent factors. The partial correlation between factors i and j, holding all other factors fixed, is

$$\rho_{ij} = -\frac{K_{ij}}{\sqrt{K_{ii}\,K_{jj}}}. \tag{F.1}$$

Each off-diagonal entry of the resulting matrix is the conditional association between two factors net of every other factor in the solution. Under standard multivariate-Gaussian assumptions, $\rho_{ij} = 0$ corresponds to conditional independence of F_i and F_j given the remaining factors; the partial-correlation network is the Gaussian graphical model implied by Φ (Epskamp and Fried, 2018). Equation (F.1) is a deterministic transform of Φ and introduces no estimation step beyond the factor analysis itself. Figure F.6 applies the same edge-retention rule used in Figure F.5 ($|\rho_{ij}| \geq 0.10$) so that the two graphs are visually comparable.

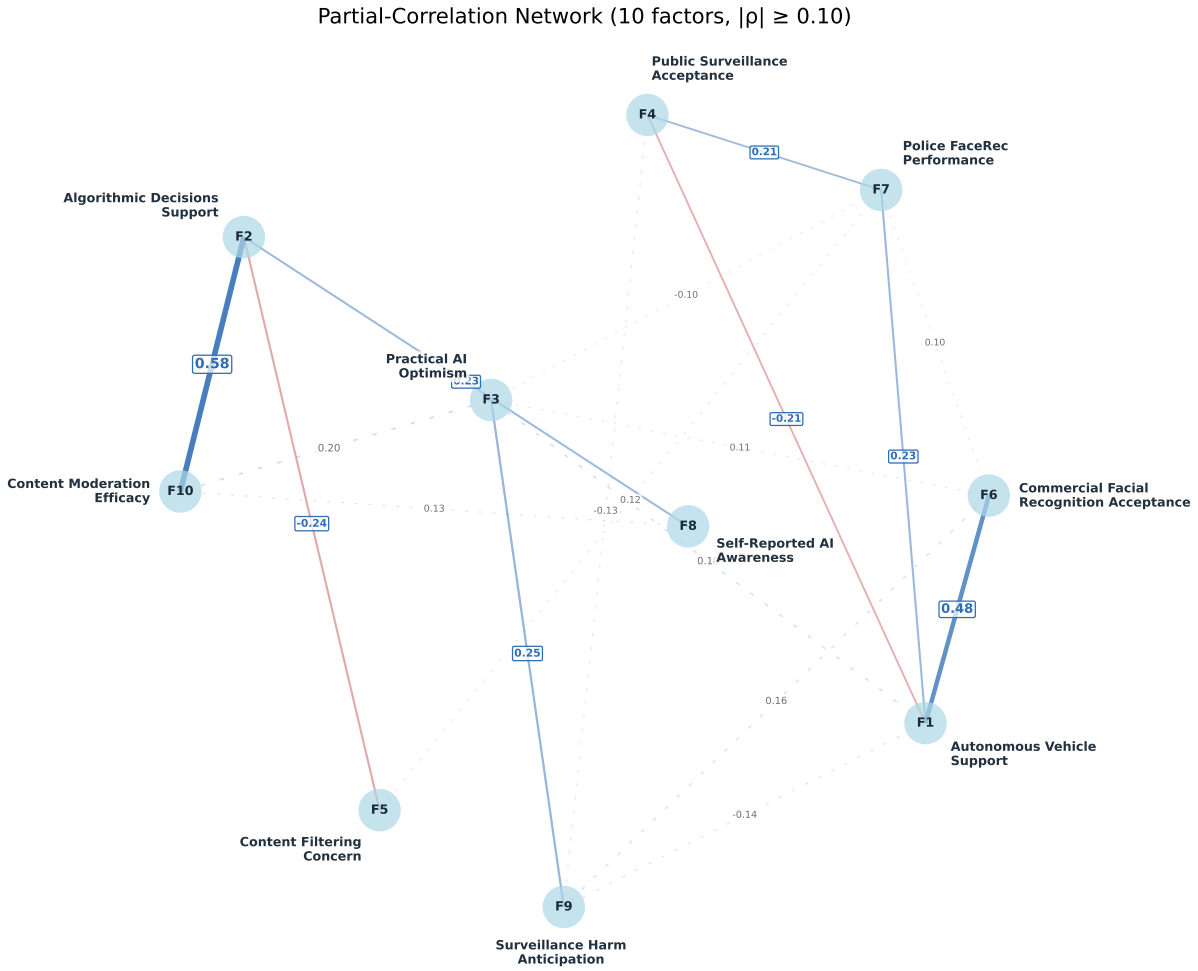


Figure F.6: Ten-factor partial-correlation network at $|\rho_{ij}| \geq 0.10$. Blue edges are positive partial correlations, red edges negative; edge thickness scales with $|\rho_{ij}|$. Edges represent conditional associations between two factors with all other factors held constant.

What the partial-correlation network adds to Figure F.5. Three substantive differences follow from conditioning on the remaining factors.

First, the dense web of weak ties in the bivariate graph thins markedly under partialing. Many of the small raw correlations are recovered as conditional independencies, indicating that those factor pairs were communicating through other factors rather than maintaining a direct association of their own.

Second, a small backbone of conditional dependencies survives. The strongest positive partial edges connect Algorithmic Decisions Support (F_2) to Content Moderation Efficacy (F_{10}) at $\rho = 0.58$ and Autonomous Vehicle Support (F_1) to Commercial Facial Recognition Acceptance (F_6) at $\rho = 0.47$. The strongest negative edges link Algorithmic Decisions Support (F_2) to Content Filtering Concern (F_5) at $\rho = -0.24$ and Autonomous Vehicle Support (F_1) to Public Surveillance Acceptance (F_4) at $\rho = -0.21$. These ties identify factor pairs whose link is not reducible to shared third-factor variance.

Third, because indirect propagation through other factors has been partialed out by construction, the $1/|\rho_{ij}|$ distance is interpretable in the conventional graph-theoretic sense within this network. Edge absence in the partial network corresponds to conditional independence given all other factors and is therefore structurally informative.

Table F.4: Node centrality metrics for the 10-factor partial-correlation network (edges retained at $|p_{ij}| \geq 0.10$).

Factor	Label	Degree	Strength	Closeness	Betweenness	Avg. Path Length
F_1	Autonomous Vehicle Support	5	1.233	0.118	0.333	8.502
F_2	Algorithmic Decisions Support	3	1.045	0.107	0.333	9.380
F_{10}	Content Moderation Efficacy	3	0.911	0.111	0.333	8.997
F_6	Commercial Facial Recognition Acceptance	4	0.856	0.100	0.000	10.034
F_3	Practical AI Optimism	5	0.846	0.126	0.361	7.926
F_7	Police FaceRec Performance	5	0.763	0.101	0.083	9.946
F_9	Surveillance Harm Anticipation	4	0.687	0.105	0.000	9.546
F_4	Public Surveillance Acceptance	3	0.549	0.088	0.000	11.331
F_8	Self-Reported AI Awareness	2	0.358	0.075	0.000	13.297
F_5	Content Filtering Concern	2	0.356	0.096	0.083	10.460

Table F.5: Ten most structurally distant factor pairs in the 10-factor partial-correlation network, sorted by shortest-path distance descending. *Direct distance* equals $1/|p_{ij}|$ when factors i and j share a retained edge ($|p_{ij}| \geq 0.10$); undefined otherwise. *Shortest-path distance* is the Dijkstra path length on the $1/|p_{ij}|$ edge-weighted graph, valid here because partial correlations have already removed the multi-hop confound that applies to bivariate r.

Factor 1	Factor 2	Direct distance	Shortest-path distance	Direct edge
F_4 (Public Surveillance Acceptance)	F_8 (Self-Reported AI Awareness)	–	21.560	No
F_6 (Commercial Facial Recognition Acceptance)	F_8 (Self-Reported AI Awareness)	–	18.860	No
F_2 (Algorithmic Decisions Support)	F_4 (Public Surveillance Acceptance)	–	17.154	No
F_7 (Police FaceRec Performance)	F_8 (Self-Reported AI Awareness)	–	17.112	No
F_1 (Autonomous Vehicle Support)	F_8 (Self-Reported AI Awareness)	–	16.755	No
F_{10} (Content Moderation Efficacy)	F_4 (Public Surveillance Acceptance)	–	15.430	No
F_8 (Self-Reported AI Awareness)	F_9 (Surveillance Harm Anticipation)	–	15.110	No
F_5 (Content Filtering Concern)	F_6 (Commercial Facial Recognition Acceptance)	–	14.960	No

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Table F.5 continued

Factor 1	Factor 2	Direct distance	Shortest-path distance	Direct edge
F_5 (Content Filtering Concern)	F_9 (Surveillance Harm Anticipation)	–	14.902	No
F_2 (Algorithmic Decisions Support)	F_6 (Commercial Facial Recognition Acceptance)	–	14.454	No

Note. F_1: Autonomous Vehicle Support; F_2: Algorithmic Decisions Support; F_3: Practical AI Optimism; F_4: Public Surveillance Acceptance; F_5: Content Filtering Concern; F_6: Commercial Facial Recognition Acceptance; F_7: Police FaceRec Performance; F_8: Self-Reported AI Awareness; F_9: Surveillance Harm Anticipation; F_{10}: Content Moderation Efficacy. Higher scores reflect more pro-AI / accepting attitudes for F_1, F_2, F_3, F_4, F_6, F_7, F_{10}, and more anti-AI / skeptical attitudes for F_5 (concern about content-filtering harms) and F_9 (anticipation of surveillance harms). F_8 is a knowledge dimension with no attitudinal valence.

Appendix F.3. Belief-System Network (Regularized Partial Correlations)

The main-text Figure 1 estimates the belief-system network of the ten latent factors using EBICglasso (Epskamp and Fried, 2018), the regularized partial-correlation estimator standard in the psychometric-network literature (Brandt et al., 2019; Brandt, 2020; Turner-Zwinkels and Brandt, 2022). The procedure applies a graphical-lasso shrinkage to the inverse of Φ and selects the tuning parameter by extended Bayesian information criterion with hyperparameter $\gamma = 0.5$ (Foygel and Drton, 2010). Edges with negligible partial correlations are shrunk exactly to zero, so the recovered network is sparser than the unregularized partial graph in Appendix F.2.

Edge-weight and centrality uncertainty are quantified with a parametric bootstrap: at each of $B = 1{,}000$ replicates, a sample of size $N = 5{,}153$ is drawn from $\text{MVN}(0, \Phi)$, EBICglasso is refit, and the recovered edges and centralities are recorded. Bootstrap confidence intervals follow the conventions of Epskamp et al. (2018); the centrality-stability coefficient (CS) summarizes how much of the original sample can be discarded while maintaining a correlation ≥ 0.70 with the original centrality ordering, with CS ≥ 0.50 taken as adequate stability and ≥ 0.25 as the minimum interpretable threshold. We further report node predictability (R^2 of each factor regressed on its neighbors) following Haslbeck and Waldorp (2018). The bootstrap quantifies sampling variability of Φ at the latent level under $\text{MVN}(0, \Phi)$; uncertainty arising from EFA extraction itself is not propagated, a limitation we acknowledge but do not attempt to address within the current sample.

Table F.6 lists the strongest partial-correlation edges with bootstrap 95% intervals; Table F.7 reports the standard centrality metrics plus predictability; Table F.8 reports the CS-coefficient by metric. Figure F.7 shows the case-drop stability curves on which the CS-coefficients are based: all four centrality metrics are stable well above the adequacy threshold (CS ≥ 0.50), with strength and expected influence retaining their ordering even when 95% of cases are dropped (CS = 0.95) and closeness and betweenness when 80% are dropped (CS = 0.80).

Table F.6: Top partial-correlation edges in the belief-system network, with 95% CIs from a parametric bootstrap ($B = 1{,}000$ draws from $\text{MVN}(0, \Phi)$).

Node 1	Node 2	Partial *r*	CI low	CI high
F2 (Algorithmic Decisions Support)	F10 (Content Moderation Efficacy)	0.590	0.572	0.607
F1 (Autonomous Vehicle Support)	F6 (Commercial Facial Recognition Acceptance)	0.484	0.464	0.504

Continued on next page

Table F.6 continued

Node 1	Node 2	Partial *r*	CI low	CI high
F3 (Practical AI Optimism)	F9 (Surveillance Harm Anticipation)	0.244	0.219	0.270
F1 (Autonomous Vehicle Support)	F7 (Police FaceRec Performance)	0.232	0.207	0.258
F2 (Algorithmic Decisions Support)	F5 (Content Filtering Concern)	−0.224	−0.251	−0.197
F2 (Algorithmic Decisions Support)	F8 (Self-Reported AI Awareness)	0.203	0.177	0.229
F3 (Practical AI Optimism)	F10 (Content Moderation Efficacy)	0.202	0.180	0.224
F1 (Autonomous Vehicle Support)	F4 (Public Surveillance Acceptance)	−0.196	−0.223	−0.169
F4 (Public Surveillance Acceptance)	F7 (Police FaceRec Performance)	0.193	0.165	0.221
F1 (Autonomous Vehicle Support)	F3 (Practical AI Optimism)	0.185	0.159	0.211
F6 (Commercial Facial Recognition Acceptance)	F9 (Surveillance Harm Anticipation)	0.163	0.136	0.190
F1 (Autonomous Vehicle Support)	F9 (Surveillance Harm Anticipation)	−0.143	−0.173	−0.113
F8 (Self-Reported AI Awareness)	F10 (Content Moderation Efficacy)	0.141	0.115	0.167
F4 (Public Surveillance Acceptance)	F9 (Surveillance Harm Anticipation)	−0.133	−0.161	−0.106
F4 (Public Surveillance Acceptance)	F5 (Content Filtering Concern)	−0.115	−0.141	−0.088
F3 (Practical AI Optimism)	F6 (Commercial Facial Recognition Acceptance)	0.103	0.076	0.131
F5 (Content Filtering Concern)	F7 (Police FaceRec Performance)	0.098	0.072	0.125
F6 (Commercial Facial Recognition Acceptance)	F7 (Police FaceRec Performance)	0.093	0.065	0.120
F3 (Practical AI Optimism)	F7 (Police FaceRec Performance)	−0.087	−0.114	−0.060
F4 (Public Surveillance Acceptance)	F8 (Self-Reported AI Awareness)	0.078	0.050	0.107

Note. F_1: Autonomous Vehicle Support; F_2: Algorithmic Decisions Support; F_3: Practical AI Optimism; F_4: Public Surveillance Acceptance; F_5: Content Filtering Concern; F_6: Commercial Facial Recognition Acceptance; F_7: Police FaceRec Performance; F_8: Self-Reported AI Awareness; F_9: Surveillance Harm Anticipation; F_{10}: Content Moderation Efficacy. Higher scores reflect more pro-AI / accepting attitudes for F_1, F_2, F_3, F_4, F_6, F_7, F_{10}, and more anti-AI / skeptical attitudes for F_5 (concern about content-filtering harms) and F_9 (anticipation of surveillance harms). F_8 is a knowledge dimension with no attitudinal valence.

Table F.7: Centrality metrics and node predictability (R^2) for the belief-system network, sorted by strength.

Factor	Label	Betweenness	Closeness	Strength	EI	Predictability (R^2)
F1	Autonomous Vehicle Support	11	0.013	1.296	0.617	0.426
F2	Algorithmic Decisions Support	10	0.012	1.102	0.603	0.529

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Table F.7 continued

Factor	Label	Betweenness	Closeness	Strength	EI	Predictability (R^2)
F10	Content Moderation Efficacy	10	0.012	1.065	0.948	0.522
F6	Commercial Facial Recognition Acceptance	0	0.011	0.979	0.910	0.402
F3	Practical AI Optimism	11	0.014	0.889	0.580	0.271
F4	Public Surveillance Acceptance	3	0.012	0.822	−0.278	0.148
F9	Surveillance Harm Anticipation	0	0.012	0.808	0.188	0.158
F7	Police FaceRec Performance	0	0.011	0.805	0.486	0.152
F8	Self-Reported AI Awareness	0	0.009	0.596	0.515	0.184
F5	Content Filtering Concern	3	0.011	0.509	−0.169	0.111

Strength: sum of absolute edge weights. Closeness: inverse of average shortest path distance. Betweenness: number of shortest paths passing through the node. EI: 1-step expected influence (sum of signed edge weights). Predictability is the variance in each node explained by all other nodes.

Table F.8: Centrality-stability coefficients (CS-coefficient, case-drop bootstrap, $B = 1000$).

Centrality metric	CS coefficient	Stable
betweenness	0.800	Yes
closeness	0.800	Yes
expectedInfluence	0.950	Yes
strength	0.950	Yes

CS-coefficient is the maximum proportion of cases that can be dropped to retain a 0.70 correlation with the original centrality estimates with 95% probability. Values ≥ 0.50 indicate adequate stability; ≥ 0.25 is the minimum interpretable threshold (Epskamp, Borsboom & Fried, 2018).

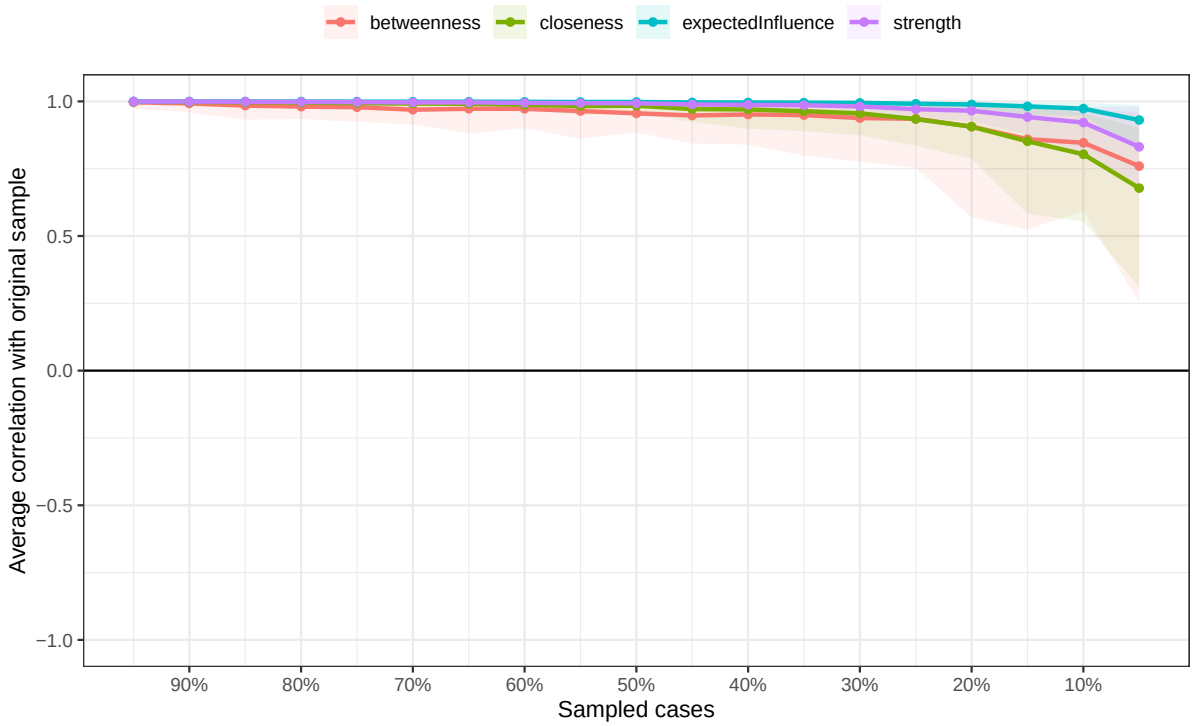


Figure F.7: Centrality stability under case-drop bootstrap (B = 1,000). Lines show the average correlation between the original centrality ordering and the centrality recovered after dropping a fraction of cases, with 95% bootstrap bands. Strength and expected influence stay stable across nearly the full case-drop range (CS = 0.95). Closeness and betweenness decline only in the smallest subsamples (CS = 0.80; Table F.8).

Appendix G. Factor Score Distributions

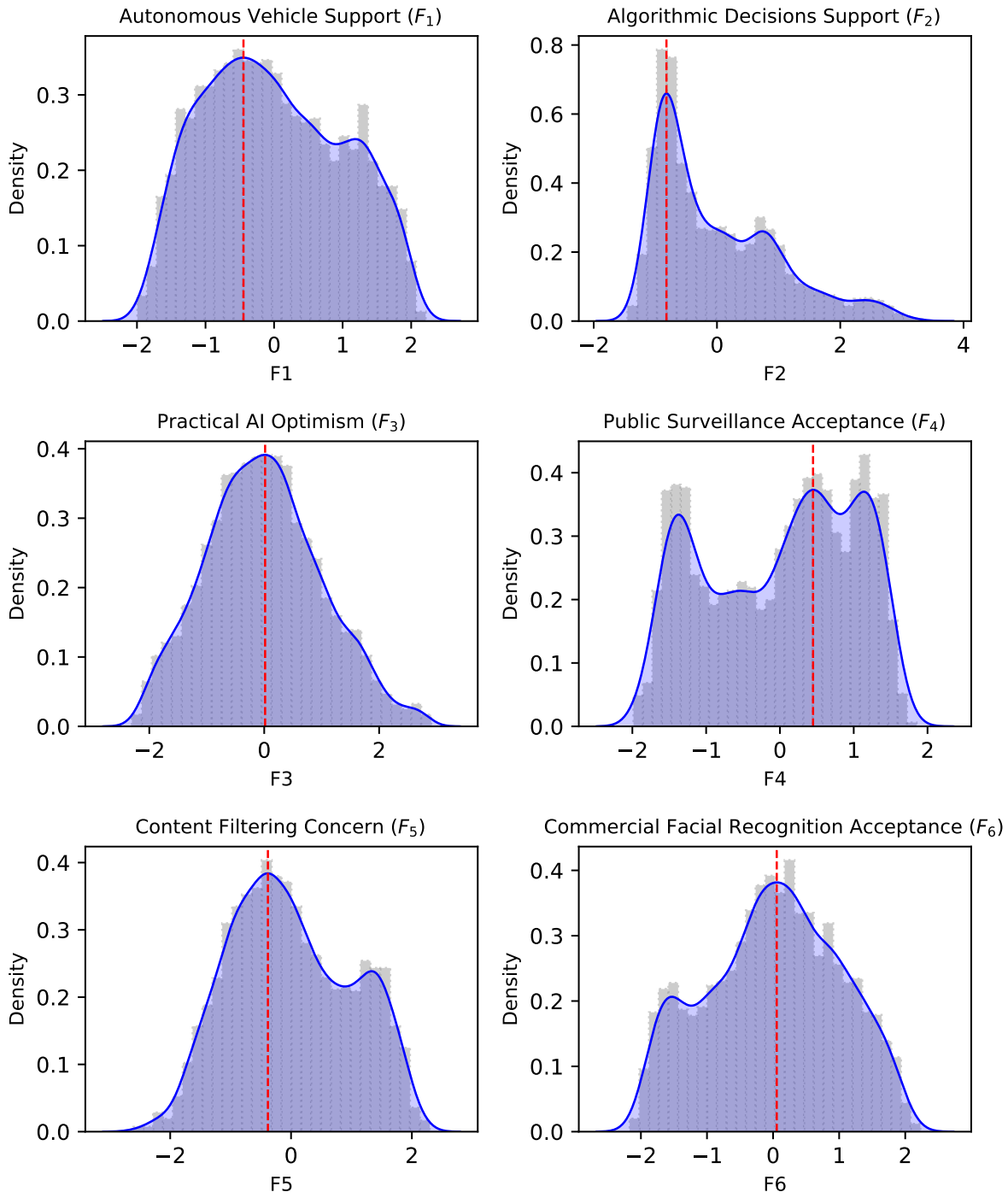


Figure G.8: Standardized factor score distributions for factors F1–F6.

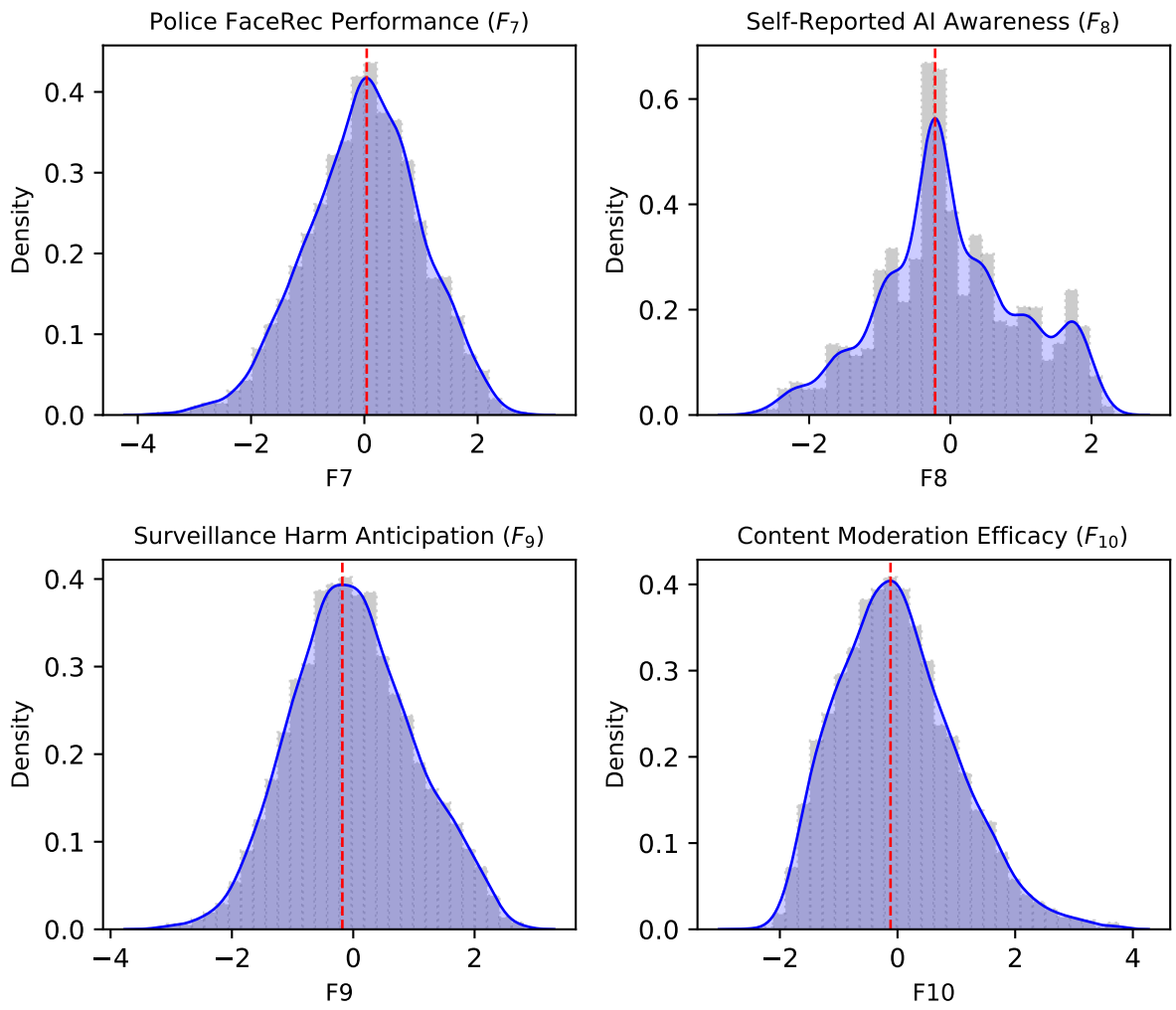


Figure G.9: Standardized factor score distributions for factors F7–F10.

Reading Figures G.8 and G.9 against Table G.9, the ten distributions are predominantly unimodal, though two cross the bimodality threshold.[4] Bimodality coefficients range from

[4] *Skew* is the Fisher–Pearson sample skewness; *Excess kurt.* is excess kurtosis (Fisher convention). *BC* is Pearson's bimodality coefficient; values exceeding $5/9 \approx 0.556$ (marked †) are consistent with a bimodal underlying distribution (Pfister et al., 2013). *Entropy* is Shannon entropy on 30 equal-width bins, normalized by

0.35 to 0.61, and two factors cross Pearson's $5/9 \approx 0.556$ cutoff: F_2 (Algorithmic Decisions Support, BC = 0.61) and F_4 (Public Surveillance Acceptance, BC = 0.59), each consistent with a modestly split underlying distribution rather than a single dominant mode. F_2 (Algorithmic Decisions Support) and F_{10} (Content Moderation Efficacy) are the most right-skewed (skewness 0.96 and 0.51), with most respondents concentrated at the lower, less-supportive end and a thin upper tail of strongly supportive respondents, whereas F_8 (Self-Reported AI Awareness), F_6 (Commercial Facial Recognition Acceptance), and F_9 (Surveillance Harm Anticipation) are near-symmetric (skewness within ±0.11). Agreement varies more widely across factors (A between 0.04 and 0.57): consensus is tightest on F_7 (Police FaceRec Performance, $A = 0.57$) and F_{10} (Content Moderation Efficacy, $A = 0.56$) and opinion most dispersed on F_4 (Public Surveillance Acceptance, $A = 0.04$), the factor that also registers the lower of the two bimodal coefficients.

Table G.9: Shape statistics for the ten z-standardized factor-score distributions ($N = 5,153$ respondents). Column definitions are given in the text preceding this table; values marked † exceed Pearson's bimodality threshold $5/9 \approx 0.556$.

Factor	Skew	Excess kurt.	BC	Entropy	A
F_1	0.18	−0.99	0.51	0.97	0.23
F_2	0.96	0.18	0.61†	0.88	0.46
F_3	0.23	−0.29	0.39	0.93	0.42
F_4	−0.21	−1.24	0.59†	0.96	0.04
F_5	0.15	−0.83	0.47	0.93	0.38
F_6	−0.09	−0.82	0.46	0.96	0.27
F_7	−0.21	−0.05	0.35	0.86	0.57
F_8	0.05	−0.33	0.37	0.91	0.42
F_9	0.11	−0.27	0.37	0.88	0.53
F_{10}	0.51	0.13	0.40	0.87	0.56

Note. F_1: Autonomous Vehicle Support; F_2: Algorithmic Decisions Support; F_3: Practical AI Optimism; F_4: Public Surveillance Acceptance; F_5: Content Filtering Concern; F_6: Commercial Facial Recognition Acceptance; F_7: Police FaceRec Performance; F_8: Self-Reported AI Awareness; F_9: Surveillance Harm Anticipation; F_{10}: Content Moderation Efficacy. Higher scores reflect more pro-AI / accepting attitudes for F_1, F_2, F_3, F_4, F_6, F_7, F_{10}, and more anti-AI / skeptical attitudes for F_5 (concern about content-filtering harms) and F_9 (anticipation of surveillance harms). F_8 is a knowledge dimension with no attitudinal valence.

Appendix H. Regression Predictor Summary

Table H.10: Descriptive statistics for the operationalized regression predictors.

Feature	Operationalization	Role	n	Unw. %	Wtd. %	M	SD	Range
Education	Less than high school	Ref.	111	2.4	5.6	–	–	–
Education	HS graduate	Ind.	613	13.4	27.9	–	–	–
Education	Some college	Ind.	940	20.6	21.0	–	–	–
Education	Associate's	Ind.	501	11.0	10.9	–	–	–
Education	College graduate	Ind.	1,306	28.6	19.7	–	–	–

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log(30) so 0 is a point mass and 1 is uniform. A is the Van der Eijk agreement statistic computed on 7 ordered bins (van der Eijk, 2001): +1 indicates full consensus on a single bin, 0 a uniform distribution, and −1 a fully bipolar split.

Table H.10 continued

Feature	Operationalization	Role	n	Unw. %	Wtd. %	M	SD	Range
Education	Postgraduate	Ind.	1,097	24.0	14.9	–	–	–
Party	Rep/Lean Rep	Ref.	2,199	48.1	45.7	–	–	–
Party	Dem/Lean Dem	Ind.	2,369	51.9	54.3	–	–	–
Age	18-29	Ref.	419	9.2	19.5	–	–	–
Age	30-49	Ind.	1,517	33.2	36.0	–	–	–
Age	50-64	Ind.	1,332	29.2	24.3	–	–	–
Age	65+	Ind.	1,300	28.5	20.1	–	–	–
Gender	Man	Ref.	2,079	45.5	48.6	–	–	–
Gender	Woman	Ind.	2,489	54.5	51.4	–	–	–
Race/ethnicity	White non-Hispanic	Ref.	3,262	71.4	64.8	–	–	–
Race/ethnicity	Black	Ind.	344	7.5	10.8	–	–	–
Race/ethnicity	Hispanic	Ind.	648	14.2	15.4	–	–	–
Race/ethnicity	Other	Ind.	149	3.3	2.9	–	–	–
Race/ethnicity	Asian	Ind.	165	3.6	6.1	–	–	–
Income	Lower income tier	Ref.	939	20.6	29.0	–	–	–
Income	Middle income tier	Ind.	2,437	53.3	52.3	–	–	–
Income	Upper income tier	Ind.	1,192	26.1	18.8	–	–	–
Religion	Protestant	Ind.	1,945	42.6	40.6	–	–	–
Religion	Catholic	Ind.	956	20.9	21.2	–	–	–
Religion	Unaffiliated	Ref.	1,284	28.1	30.1	–	–	–
Religion	Other	Ind.	383	8.4	8.1	–	–	–
Region	Northeast	Ref.	741	16.2	17.5	–	–	–
Region	Midwest	Ind.	995	21.8	20.9	–	–	–
Region	South	Ind.	1,789	39.2	38.0	–	–	–
Region	West	Ind.	1,043	22.8	23.6	–	–	–
Zone	Metropolitan	Ref.	4,034	88.3	87.5	–	–	–
Zone	Non-metropolitan	Ind.	534	11.7	12.5	–	–	–
Marital status	Married	Ind.	2,653	58.1	53.0	–	–	–
Marital status	Living with partner	Ind.	370	8.1	9.8	–	–	–
Marital status	Divorced	Ind.	527	11.5	9.4	–	–	–
Marital status	Separated	Ind.	90	2.0	2.3	–	–	–
Marital status	Widowed	Ind.	227	5.0	3.9	–	–	–
Marital status	Never married	Ref.	701	15.3	21.6	–	–	–
Ideology	Higher = more liberal	Cont.	4,568	–	–	2.91	1.07	1.00–5.00
Service attendance	Higher = less frequent	Cont.	4,568	–	–	4.21	1.62	1.00–6.00
Internet use	Higher = less frequent	Cont.	4,568	–	–	1.74	0.85	1.00–5.00

Note. Summary statistics are computed on the shared regression estimation sample ($N = 4,568$): respondents listwise-complete on all ten factor scores and all predictors after Pew refusal codes are set to missing. Respondents describing their gender in some other way are excluded because the category is too small for a stable regression dummy. Percentages are shown both unweighted and weighted by WEIGHT_W99. Continuous-predictor means and standard deviations are weighted. Role abbreviations: Ref. = reference category, Ind. = indicator contrast, Cont. = continuous predictor.

Appendix I. Full Regression Results

Table I.11: Weighted least-squares regressions of the factor scores on the full sociodemographic and political predictor set, factors F_1–F_5.

Predictor	F_1	F_2	F_3	F_4	F_5
Intercept	0.225	0.364	0.441	0.075	1.331***
	(0.197)	(0.254)	(0.227)	(0.230)	(0.184)
Education: HS graduate	−0.141	−0.163	−0.176	−0.061	−0.106
	(0.118)	(0.140)	(0.155)	(0.115)	(0.112)
Education: Some college	−0.111	−0.370**	−0.190	−0.141	−0.085
	(0.118)	(0.141)	(0.155)	(0.115)	(0.112)
Education: Associate's	0.004	−0.360*	−0.052	−0.145	0.062
	(0.131)	(0.145)	(0.167)	(0.126)	(0.114)
Education: College graduate	0.170	−0.241	0.065	−0.248*	−0.072
	(0.120)	(0.142)	(0.155)	(0.116)	(0.111)
Education: Postgraduate	0.214	−0.265	0.153	−0.188	−0.162
	(0.123)	(0.142)	(0.158)	(0.119)	(0.111)

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Table I.11 continued

Predictor	F_1	F_2	F_3	F_4	F_5
Party: Dem/Lean Dem	0.199***	0.186*	0.146*	−0.050	−0.736***
	(0.057)	(0.078)	(0.064)	(0.066)	(0.060)
Ideology (higher = liberal)	0.093***	0.025	0.036	−0.064	−0.187***
	(0.028)	(0.040)	(0.033)	(0.035)	(0.026)
Age: 30-49	−0.100	−0.029	−0.202*	0.025	−0.073
	(0.075)	(0.103)	(0.089)	(0.090)	(0.078)
Age: 50-64	−0.373***	−0.218*	−0.463***	0.034	0.053
	(0.083)	(0.109)	(0.095)	(0.096)	(0.084)
Age: 65+	−0.326***	−0.179	−0.371***	0.264**	−0.013
	(0.090)	(0.111)	(0.102)	(0.096)	(0.087)
Gender: Woman	−0.509***	−0.182***	−0.385***	0.092*	−0.020
	(0.040)	(0.048)	(0.045)	(0.046)	(0.042)
Race/eth.: Black	−0.083	0.080	0.060	−0.115	−0.146
	(0.074)	(0.094)	(0.083)	(0.084)	(0.085)
Race/eth.: Hispanic	0.100	0.197*	0.201*	0.151	−0.052
	(0.072)	(0.090)	(0.079)	(0.086)	(0.077)
Race/eth.: Other	−0.110	−0.214	−0.044	−0.321*	0.184
	(0.107)	(0.236)	(0.187)	(0.158)	(0.147)
Race/eth.: Asian	0.067	0.187	0.261	0.124	−0.241*
	(0.097)	(0.130)	(0.156)	(0.122)	(0.116)
Income: Middle tier	−0.017	−0.254***	−0.076	0.021	−0.042
	(0.056)	(0.071)	(0.063)	(0.063)	(0.054)
Income: Upper tier	0.125	−0.293***	0.038	0.114	−0.160*
	(0.067)	(0.079)	(0.074)	(0.079)	(0.062)
Internet use (higher = less frequent)	−0.099**	−0.031	−0.082	−0.033	−0.014
	(0.033)	(0.033)	(0.044)	(0.030)	(0.029)
Religion: Protestant	−0.182**	−0.020	−0.222**	−0.084	−0.083
	(0.062)	(0.076)	(0.072)	(0.077)	(0.065)
Religion: Catholic	−0.146*	−0.010	−0.142	−0.030	−0.122
	(0.069)	(0.082)	(0.086)	(0.086)	(0.065)
Religion: Other	−0.035	0.120	0.030	0.008	0.002
	(0.082)	(0.128)	(0.095)	(0.097)	(0.087)
Service attendance (higher = less frequent)	0.019	0.032	0.047**	0.023	−0.032*
	(0.016)	(0.019)	(0.018)	(0.019)	(0.016)
Region: Midwest	0.085	−0.038	0.023	−0.041	−0.156*
	(0.064)	(0.083)	(0.082)	(0.078)	(0.070)
Region: South	0.139*	0.060	0.065	0.065	−0.051
	(0.060)	(0.084)	(0.081)	(0.073)	(0.065)
Region: West	0.110	−0.068	−0.020	−0.058	−0.156*
	(0.067)	(0.083)	(0.086)	(0.079)	(0.069)
Zone: Non-metropolitan	−0.130	0.040	−0.039	0.096	0.064
	(0.066)	(0.077)	(0.067)	(0.073)	(0.062)
Marital: Married	−0.032	0.035	−0.026	0.148	0.061
	(0.067)	(0.087)	(0.078)	(0.080)	(0.068)
Marital: Living with partner	−0.007	−0.068	−0.089	0.068	−0.024
	(0.082)	(0.108)	(0.094)	(0.097)	(0.081)
Marital: Divorced	−0.059	0.106	−0.027	0.025	0.108
	(0.083)	(0.098)	(0.087)	(0.095)	(0.092)

Marital: Separated	−0.237	0.034	−0.343	0.159	0.061
	(0.150)	(0.187)	(0.188)	(0.155)	(0.167)
Marital: Widowed	0.067	−0.059	0.052	0.059	0.079
	(0.097)	(0.127)	(0.121)	(0.113)	(0.109)
N	4,568	4,568	4,568	4,568	4,568
Adjusted R^2	0.236	0.088	0.182	0.040	0.330

Note. F_1: Autonomous Vehicle Support; F_2: Algorithmic Decisions Support; F_3: Practical AI Optimism; F_4: Public Surveillance Acceptance; F_5: Content Filtering Concern; F_6: Commercial Facial Recognition Acceptance; F_7: Police FaceRec Performance; F_8: Self-Reported AI Awareness; F_9: Surveillance Harm Anticipation; F_{10}: Content Moderation Efficacy. Higher scores reflect more pro-AI / accepting attitudes for F_1, F_2, F_3, F_4, F_6, F_7, F_{10}, and more anti-AI / skeptical attitudes for F_5 (concern about content-filtering harms) and F_9 (anticipation of surveillance harms). F_8 is a knowledge dimension with no attitudinal valence. WLS estimates (survey weight WEIGHT_W99), HC3 robust standard errors in parentheses; listwise estimation samples. Reference categories: *Education*: less than high school, *Party*: Rep/Lean Rep, *Age*: 18–29, *Gender*: man, *Race/eth.*: White non-Hispanic, *Income*: lower income tier, *Religion*: unaffiliated, *Region*: Northeast, *Zone*: metropolitan, *Marital*: never married. Respondents describing their gender in some other way are excluded (too few cases for a stable dummy). $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

Table I.12: Weighted least-squares regressions of the factor scores on the full sociodemographic and political predictor set, factors F_6–F_{10}.

Predictor	F_6	F_7	F_8	F_9	F_{10}
Intercept	0.339	0.279	0.449	−0.082	0.617*
	(0.227)	(0.244)	(0.253)	(0.229)	(0.248)
Education: HS graduate	−0.201	0.096	−0.123	0.138	−0.155
	(0.113)	(0.155)	(0.168)	(0.175)	(0.141)
Education: Some college	−0.318**	−0.017	0.002	0.321	−0.477**
	(0.114)	(0.157)	(0.164)	(0.179)	(0.145)
Education: Associate's	−0.313*	−0.018	0.115	0.077	−0.370*
	(0.123)	(0.165)	(0.171)	(0.189)	(0.148)
Education: College graduate	−0.406***	−0.035	0.156	0.293	−0.449**
	(0.110)	(0.157)	(0.163)	(0.176)	(0.143)
Education: Postgraduate	−0.383***	0.029	0.270	0.174	−0.409**
	(0.113)	(0.162)	(0.168)	(0.183)	(0.144)
Party: Dem/Lean Dem	0.129	−0.005	−0.065	0.200**	0.333***
	(0.068)	(0.068)	(0.067)	(0.076)	(0.069)
Ideology (higher = liberal)	−0.086*	−0.058	0.013	0.113**	0.035
	(0.035)	(0.040)	(0.037)	(0.041)	(0.037)
Age: 30-49	0.088	0.005	−0.054	−0.022	−0.218*
	(0.096)	(0.104)	(0.102)	(0.111)	(0.097)
Age: 50-64	0.207*	0.155	0.042	0.000	−0.391***
	(0.102)	(0.107)	(0.105)	(0.110)	(0.097)
Age: 65+	0.412***	0.298**	0.130	−0.123	−0.337***
	(0.101)	(0.107)	(0.118)	(0.113)	(0.099)
Gender: Woman	−0.018	0.047	−0.521***	−0.085	−0.092*
	(0.046)	(0.051)	(0.050)	(0.053)	(0.046)
Race/eth.: Black	−0.039	−0.318**	−0.085	0.248*	0.318***
	(0.083)	(0.102)	(0.103)	(0.114)	(0.096)
Race/eth.: Hispanic	0.171	0.180	0.083	0.299**	0.412***
	(0.088)	(0.096)	(0.092)	(0.101)	(0.086)
Race/eth.: Other	−0.128	−0.045	0.220*	0.534***	−0.054
	(0.165)	(0.198)	(0.111)	(0.159)	(0.260)
Race/eth.: Asian	0.271*	0.149	0.132	0.188	0.398**
	(0.122)	(0.168)	(0.154)	(0.199)	(0.137)
Income: Middle tier	−0.001	0.023	0.112	−0.039	−0.197**
	(0.065)	(0.069)	(0.070)	(0.074)	(0.066)
Income: Upper tier	0.034	0.015	0.162*	−0.209*	−0.182*
	(0.080)	(0.085)	(0.074)	(0.083)	(0.074)
Internet use (higher = less frequent)	−0.123***	−0.158***	−0.123**	−0.101*	−0.006
	(0.029)	(0.035)	(0.043)	(0.041)	(0.035)

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Table I.12 continued

Predictor	F_6	F_7	F_8	F_9	F_{10}
Religion: Protestant	0.021	−0.098	−0.361***	−0.078	−0.111
	(0.075)	(0.088)	(0.080)	(0.093)	(0.078)
Religion: Catholic	0.064	0.015	−0.289**	−0.222*	−0.007
	(0.082)	(0.097)	(0.092)	(0.108)	(0.084)
Religion: Other	0.098	0.025	0.037	−0.137	0.220*
	(0.094)	(0.105)	(0.112)	(0.121)	(0.101)
Service attendance (higher = less frequent)	0.020	−0.022	−0.017	−0.020	−0.008
	(0.019)	(0.022)	(0.019)	(0.022)	(0.018)
Region: Midwest	−0.044	−0.008	−0.096	−0.023	−0.049
	(0.077)	(0.082)	(0.082)	(0.084)	(0.078)
Region: South	0.067	0.122	0.045	−0.052	0.030
	(0.075)	(0.082)	(0.081)	(0.087)	(0.079)
Region: West	−0.084	0.005	−0.016	−0.091	−0.094
	(0.079)	(0.095)	(0.088)	(0.101)	(0.081)
Zone: Non-metropolitan	0.054	0.028	−0.081	−0.062	0.025
	(0.071)	(0.073)	(0.070)	(0.075)	(0.070)
Marital: Married	0.039	0.071	0.079	−0.094	−0.038
	(0.084)	(0.091)	(0.084)	(0.095)	(0.087)
Marital: Living with partner	0.045	0.040	0.099	−0.002	−0.080
	(0.096)	(0.114)	(0.123)	(0.134)	(0.099)
Marital: Divorced	−0.079	0.038	0.167	−0.019	0.017
	(0.101)	(0.116)	(0.111)	(0.106)	(0.102)
Marital: Separated	0.216	0.088	0.187	−0.176	0.052
	(0.148)	(0.167)	(0.184)	(0.152)	(0.147)
Marital: Widowed	−0.090	0.023	0.170	−0.096	0.057
	(0.118)	(0.129)	(0.127)	(0.128)	(0.119)
N	4,568	4,568	4,568	4,568	4,568
Adjusted R^2	0.045	0.047	0.138	0.114	0.190

Note. F_1: Autonomous Vehicle Support; F_2: Algorithmic Decisions Support; F_3: Practical AI Optimism; F_4: Public Surveillance Acceptance; F_5: Content Filtering Concern; F_6: Commercial Facial Recognition Acceptance; F_7: Police FaceRec Performance; F_8: Self-Reported AI Awareness; F_9: Surveillance Harm Anticipation; F_{10}: Content Moderation Efficacy. Higher scores reflect more pro-AI / accepting attitudes for F_1, F_2, F_3, F_4, F_6, F_7, F_{10}, and more anti-AI / skeptical attitudes for F_5 (concern about content-filtering harms) and F_9 (anticipation of surveillance harms). F_8 is a knowledge dimension with no attitudinal valence. WLS estimates (survey weight WEIGHT_W99), HC3 robust standard errors in parentheses; listwise estimation samples. Reference categories: *Education*: less than high school, *Party*: Rep/Lean Rep, *Age*: 18–29, *Gender*: man, *Race/eth.*: White non-Hispanic, *Income*: lower income tier, *Religion*: unaffiliated, *Region*: Northeast, *Zone*: metropolitan, *Marital*: never married. Respondents describing their gender in some other way are excluded (too few cases for a stable dummy). $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

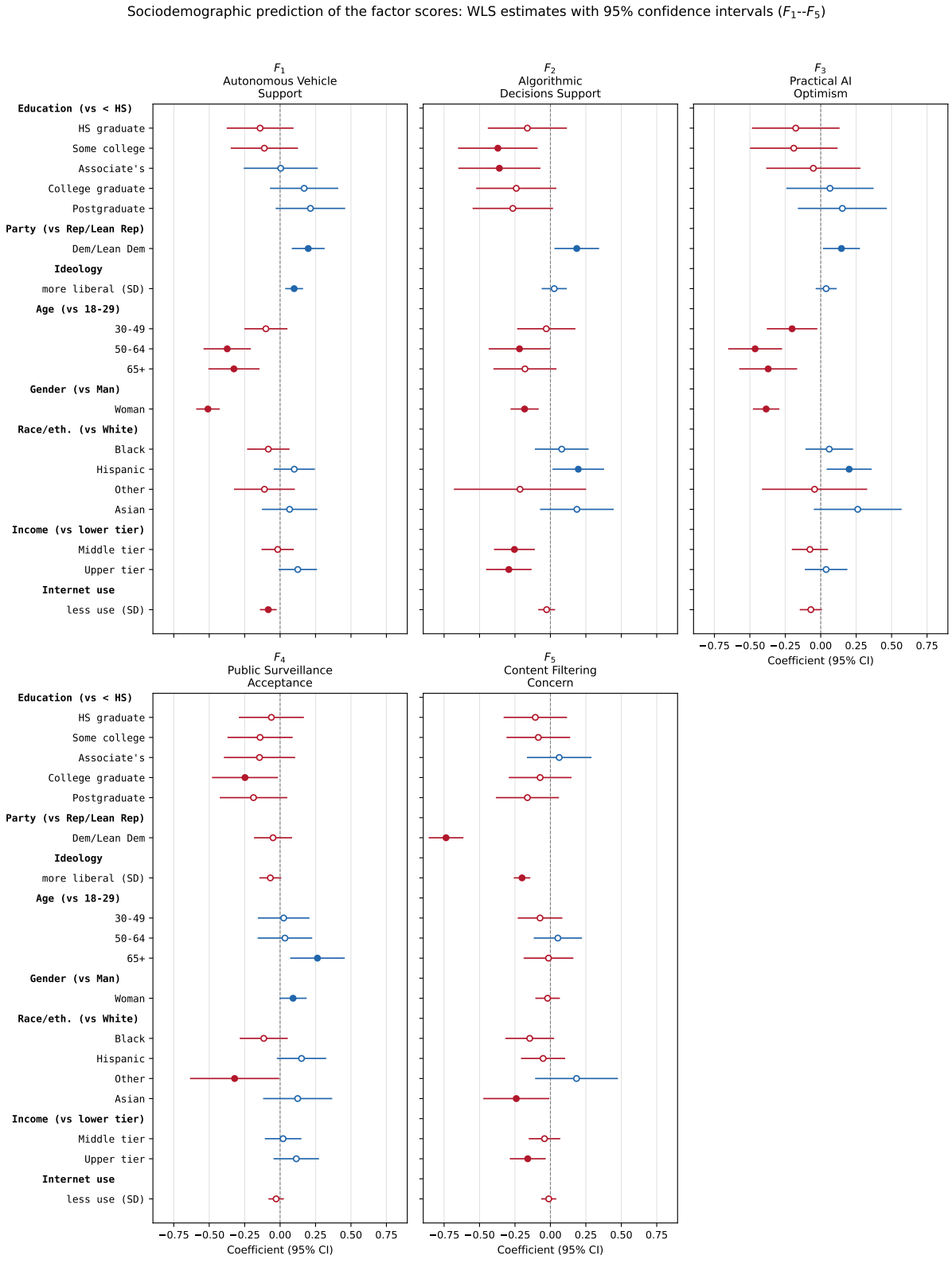


Figure I.10: Weighted least-squares estimates with 95% confidence intervals (HC3 standard errors) for factor scores F_1–F_5 (N = 4,568 per model; respondents describing their gender in some other way are excluded). The figure omits religion, service attendance, region, metropolitan status, and marital status, which are estimated but not plotted (full estimates in Tables I.11 and I.12). Coefficients for the continuous predictors shown (ideology, internet use) are scaled per standard deviation; dummy contrasts are raw differences relative to the reference category given in parentheses. Outcome factor scores are z-standardized. Blue markers denote positive estimates, red markers negative; filled markers $p < .05$. The horizontal scale is shared with Figure I.11.

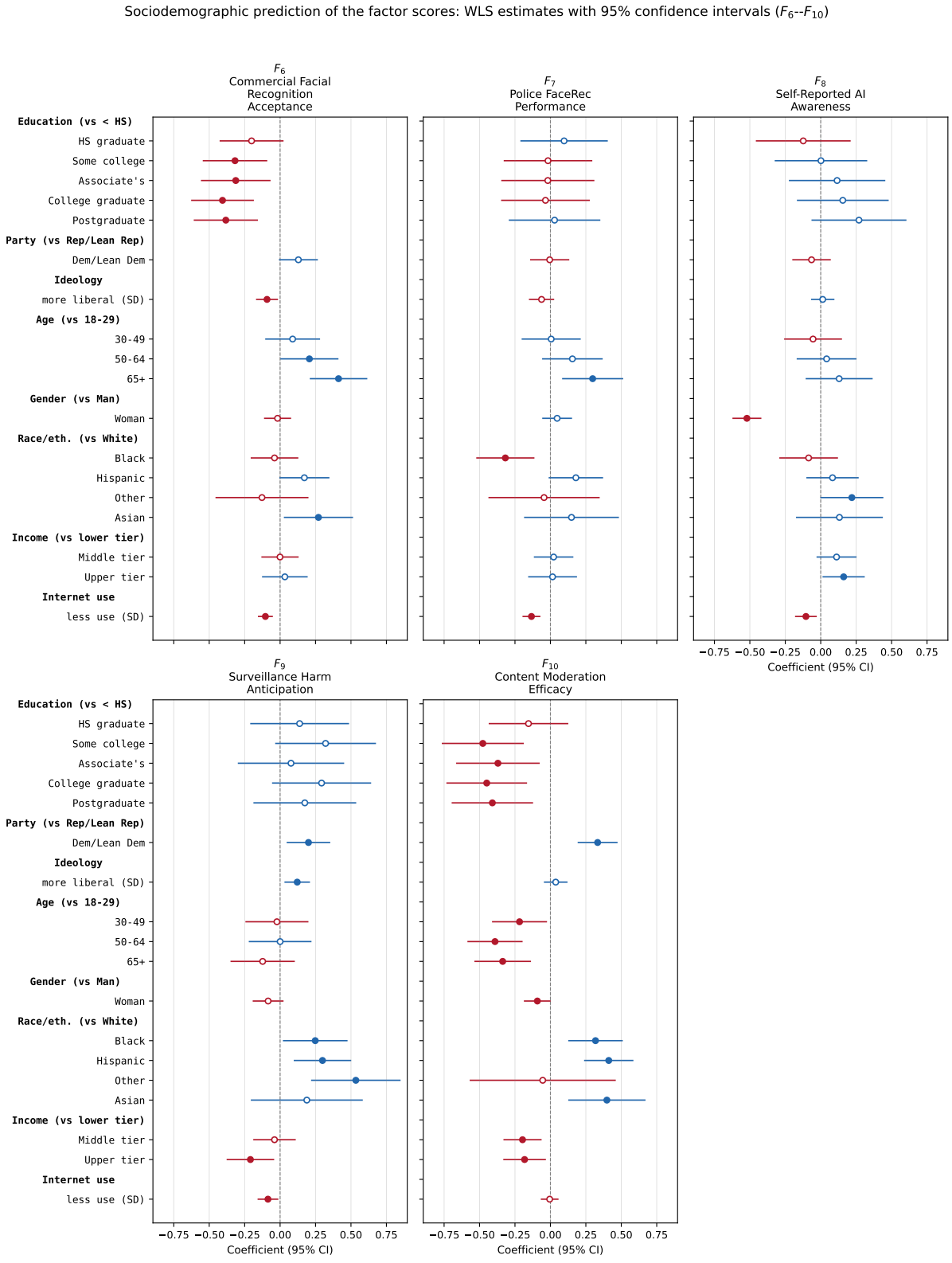


Figure I.11: Weighted least-squares estimates with 95% confidence intervals (HC3 standard errors) for factor scores F_6–F_{10}; specification, scaling, and sample as in Figure I.10.

Appendix J. General-Factor and Cross-Equation Checks

Three analyses assess claims made in the main text. First, a second-order factor analysis of the oblimin factor-correlation matrix Φ tests whether a single general factor sits above the ten first-order factors: the first eigenvalue of Φ accounts for 26.6% of the trace, and the one-factor solution absorbs 19.8% of factor-level variance (ECV), with heterogeneous loadings — moderate for the algorithmic and moderation factors, near zero for police facial-recognition performance, and negative for public-surveillance acceptance and filtering concern (Table J.13). The table's ω_h-like summary (0.552) treats the ten factors as indicators of one general factor, so it describes a unit-weighted composite of all ten factors, of which the general factor would account for about half the variance; it does not bear on how much of any single factor is general. Second, stacked-equation Wald tests formalize the cross-equation coefficient contrasts discussed in the main text: each pair of factor equations is stacked, every predictor is interacted with an equation indicator, and the model is estimated by weighted least squares with standard errors clustered on respondent (Table J.14). Every contrast the main text describes as a reversal of raw coefficient signs is significant at $p < .001$. Two rows of the table are not significant (ideology on F_1 versus F_9, and Black respondents on F_9 versus F_{10}); in both the raw slopes agree in sign and the contrast is one of valence rather than direction, which is why the text does not present them as reversals. Third, Table J.15 reports the general-index regressions referenced in the Discussion: the 52 items are collapsed into a single index (first principal component, and a valence-aligned mean) and regressed on the same predictor battery with the same weights and standard errors as the factor-score models.

Table J.13: Second-order general-factor check on the 10-factor correlation matrix

Factor	Label	Second-order loading
F_1	Autonomous Vehicle Support	0.425
F_2	Algorithmic Decisions Support	0.665
F_3	Practical AI Optimism	0.495
F_4	Public Surveillance Acceptance	−0.204
F_5	Content Filtering Concern	−0.201
F_6	Commercial Facial Recognition Acceptance	0.510
F_7	Police FaceRec Performance	0.040
F_8	Self-Reported AI Awareness	0.401
F_9	Surveillance Harm Anticipation	0.288
F_{10}	Content Moderation Efficacy	0.724
Eigenvalues of Φ: 2.66, 1.66, 1.30, 0.92, 0.87, 0.70, 0.65, 0.56, 0.38, 0.30		
First eigenvalue share of trace: 0.266; ECV: 0.198; ω_h-like: 0.552		

Note. Loadings from a one-factor minres solution fit to the 10-factor oblimin factor-correlation matrix Φ (a second-order check on Φ, not a bifactor model estimated on the 52 items). ECV is the sum of squared second-order loadings divided by $\mathrm{tr}(\Phi) = 10$, the total variance of the 10 standardized first-order factors. The ω_h-like summary treats the 10 first-order factors as indicators of the general factor.

Table J.14: Cross-equation Wald tests for the headline sign-reversal contrasts in the factor-score regressions.

Predictor	**Contrast**	β_A	β_B	$\beta_B - \beta_A$	**SE**	z	p
Age 65+ (vs 18–29)	F_3 vs. F_4	−0.371	0.264	0.634	0.119	5.317	< .001
Age 65+ (vs 18–29)	F_1 vs. F_6	−0.326	0.412	0.738	0.105	7.029	< .001
Age 65+ (vs 18–29)	F_3 vs. F_6	−0.371	0.412	0.783	0.115	6.785	< .001
Age 65+ (vs 18–29)	F_3 vs. F_7	−0.371	0.298	0.669	0.119	5.621	< .001
Ideology (higher = liberal)	F_1 vs. F_9	0.093	0.113	0.020	0.042	0.478	0.633
Ideology (higher = liberal)	F_1 vs. F_6	0.093	−0.086	−0.179	0.032	−5.624	< .001
Dem/Lean Dem (vs Rep)	F_5 vs. F_4	−0.736	−0.050	0.687	0.092	7.456	< .001
Dem/Lean Dem (vs Rep)	F_5 vs. F_9	−0.736	0.200	0.937	0.087	10.795	< .001
Woman (vs man)	F_1 vs. F_4	−0.509	0.092	0.601	0.051	11.881	< .001
Black (vs White)	F_7 vs. F_{10}	−0.318	0.318	0.636	0.086	7.411	< .001
Black (vs White)	F_9 vs. F_{10}	0.248	0.318	0.070	0.132	0.531	0.596

Note. Stacked-equation Wald tests of coefficient equality across factor-score equations. For each contrast the two WLS equations (survey weight WEIGHT_W99, identical demographic and political controls) are stacked long-format with two rows per respondent, every predictor is interacted with an equation indicator, and the focal interaction coefficient estimates the cross-equation difference $\beta_B - \beta_A$; standard errors are clustered on respondent (QKEY), so the test accounts for the covariance of the two equations' errors within respondents. Equation-specific slopes reproduce the separately estimated per-factor coefficients exactly. N = 4,568 respondents (9,136 stacked rows) per test.

Table J.15: General-index cancellation demonstration: WLS regressions of two single-index collapses of the 52 AI items on the full sociodemographic specification.

Predictor	**PC1 index β**	**SE**	p	**Aligned-mean β**	**SE**	p
Age: 30-49 (vs. 18-29)	−0.101	0.083	0.224	−0.087	0.085	0.308
Age: 50-64 (vs. 18-29)	−0.317***	0.088	< .001	−0.256**	0.090	0.004
Age: 65+ (vs. 18-29)	−0.191*	0.091	0.036	−0.100	0.092	0.276
Gender: Woman (vs. Man)	−0.387***	0.041	< .001	−0.335***	0.041	< .001
Education: HS graduate (vs. Less than high school)	−0.166	0.120	0.166	−0.183	0.116	0.113
Education: Some college (vs. Less than high school)	−0.235	0.122	0.054	−0.277*	0.118	0.019
Education: Associate's (vs. Less than high school)	−0.136	0.131	0.298	−0.187	0.125	0.134
Education: College graduate (vs. Less than high school)	−0.027	0.122	0.827	−0.103	0.118	0.381
Education: Postgraduate (vs. Less than high school)	0.044	0.125	0.724	−0.016	0.120	0.894
Income: Middle income tier (vs. Lower income tier)	−0.062	0.058	0.282	−0.063	0.058	0.276
Income: Upper income tier (vs. Lower income tier)	0.059	0.073	0.417	0.064	0.074	0.386
Race/ethnicity: Black (vs. White non-Hispanic)	−0.043	0.076	0.574	−0.023	0.075	0.761
Race/ethnicity: Hispanic (vs. White non-Hispanic)	0.203**	0.070	0.004	0.204**	0.071	0.004
Race/ethnicity: Other (vs. White non-Hispanic)	−0.171	0.188	0.363	−0.218	0.194	0.261
Race/ethnicity: Asian (vs. White non-Hispanic)	0.223	0.118	0.059	0.234*	0.119	0.050

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Table J.15 continued

Predictor	PC1 index β	SE	p	Aligned-mean β	SE	p
Party: Dem/Lean Dem (vs. Rep/Lean Rep)	0.235***	0.060	< .001	0.260***	0.060	< .001
Religion: Protestant (vs. Unaffiliated)	−0.176**	0.068	0.010	−0.164*	0.067	0.014
Religion: Catholic (vs. Unaffiliated)	−0.102	0.076	0.181	−0.077	0.075	0.305
Religion: Other (vs. Unaffiliated)	0.037	0.081	0.646	0.066	0.083	0.427
Region: Midwest (vs. Northeast)	0.040	0.072	0.586	0.028	0.072	0.702
Region: South (vs. Northeast)	0.129	0.070	0.064	0.130	0.070	0.062
Region: West (vs. Northeast)	0.033	0.075	0.656	0.025	0.073	0.730
Zone: Non-metropolitan (vs. Metropolitan)	−0.056	0.065	0.387	−0.048	0.064	0.456
Marital status: Married (vs. Never married)	0.009	0.076	0.911	0.039	0.078	0.619
Marital status: Living with partner (vs. Never married)	−0.017	0.084	0.838	0.007	0.084	0.936
Marital status: Divorced (vs. Never married)	−0.023	0.086	0.791	0.001	0.087	0.987
Marital status: Separated (vs. Never married)	−0.140	0.141	0.319	−0.036	0.132	0.784
Marital status: Widowed (vs. Never married)	0.043	0.109	0.692	0.058	0.107	0.586
Ideology (higher = liberal)	0.051	0.033	0.123	0.044	0.033	0.181
Service attendance (higher = less frequent)	0.030	0.017	0.075	0.029	0.017	0.094
Internet use (higher = less frequent)	−0.110***	0.032	< .001	−0.107***	0.029	< .001
N	4,568			4,568		
R^2	0.176			0.151		

Note. WLS coefficients (survey weight WEIGHT_W99) with HC3 robust standard errors, estimated on the factor-score regression sample ($N = 4,568$; listwise deletion, respondents reporting gender in some other way excluded). Both dependent variables collapse the 52 EFA items into a single general pro/anti-AI index and are z-standardized. PC1 index: first principal component of the pairwise-complete correlation matrix of the z-scored items (19.2% of total variance), oriented so higher = pro-AI (positively correlated with F_3, Practical AI Optimism); respondents' missing items enter at the item mean. Aligned-mean index: z-scored items with negative PC1 loadings reverse-keyed (8 of 52 items flipped), then averaged over each respondent's available items. $^{*}p < .05$; $^{**}p < .01$; $^{***}p < .001$.

Appendix K. Factor-Correlation Matrix

Table K.16 reports the full oblimin factor-correlation matrix Φ over the ten latent factors. The diagonal is omitted; off-diagonal entries are the inter-factor correlations from the rotated solution. The bivariate factor-correlation graph (Figure F.5) and the unregularized partial-correlation network (Figure F.6) are both derived from this matrix; the belief-system network in the main text (Figure 1) uses it as the input correlation matrix for EBICglasso estimation.

Table K.16: Oblimin factor-correlation matrix for the 10-factor solution. This is the matrix from which Figure 1 is derived; the belief-system network applies EBICglasso regularization to Φ.

	Label	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}
F_1	Autonomous Vehicle Support	–	0.13	0.30	−0.24	0.03	0.57	0.26	0.13	0.06	0.16
F_2	Algorithmic Decisions Support	0.13	–	0.22	−0.05	−0.31	0.20	−0.03	0.40	0.11	0.68
F_3	Practical AI Optimism	0.30	0.22	–	−0.18	−0.05	0.33	−0.05	0.10	0.32	0.33
F_4	Public Surveillance Acceptance	−0.24	−0.05	−0.18	–	−0.07	−0.19	0.14	0.04	−0.18	−0.08
F_5	Content Filtering Concern	0.03	−0.31	−0.05	−0.07	–	−0.01	0.12	−0.09	0.01	−0.21
F_6	Commercial Facial Recognition Acceptance	0.57	0.20	0.33	−0.19	−0.01	–	0.19	0.13	0.22	0.24
F_7	Police FaceRec Performance	0.26	−0.03	−0.05	0.14	0.12	0.19	–	0.05	−0.05	−0.05
F_8	Self-Reported AI Awareness	0.13	0.40	0.10	0.04	−0.09	0.13	0.05	–	0.07	0.36

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Table K.16 continued

	Label	F_1	F_2	F_3	F_4	F_5	F_6	F_7	F_8	F_9	F_{10}
F_9	Surveillance Harm Anticipation	0.06	0.11	0.32	−0.18	0.01	0.22	−0.05	0.07	–	0.16
F_{10}	Content Moderation Efficacy	0.16	0.68	0.33	−0.08	−0.21	0.24	−0.05	0.36	0.16	–

Note. F_1: Autonomous Vehicle Support; F_2: Algorithmic Decisions Support; F_3: Practical AI Optimism; F_4: Public Surveillance Acceptance; F_5: Content Filtering Concern; F_6: Commercial Facial Recognition Acceptance; F_7: Police FaceRec Performance; F_8: Self-Reported AI Awareness; F_9: Surveillance Harm Anticipation; F_{10}: Content Moderation Efficacy. Higher scores reflect more pro-AI / accepting attitudes for F_1, F_2, F_3, F_4, F_6, F_7, F_{10}, and more anti-AI / skeptical attitudes for F_5 (concern about content-filtering harms) and F_9 (anticipation of surveillance harms). F_8 is a knowledge dimension with no attitudinal valence.